Clouds Determine Diffusion: A Cloud-Deterministic Diffusion Model for Progressive Cloud Removal in Optical Remote Sensing Images

Yujun Guo, Wei He, *Senior Member, IEEE*, He Huang, and Hongyan Zhang, *Senior Member, IEEE*

***Abstract*—Cloud removal is crucial for preprocessing optical remote sensing images affected by natural cloud contamination, which is highly diverse and variable. Clouds of varying thickness often occur simultaneously, where the boundary between these layers is ambiguous and hard to quantitatively define. Consequently, advanced deep learning (DL) techniques, especially diffusion models, have been employed to address complex cloud contamination. They primarily focus on conditional generation and standard Gaussian noise diffusion. Yet this paradigm fails to fully capture the inherent diversity, complexity, and variability of natural cloud contamination, nor does it align with the reconstruction process from severe to mild states, leading to limited generalization in real-world scenes. To overcome these limitations, we propose a cloud-deterministic diffusion and progressive cloud removal (CDPCR) model. It introduces a novel diffusion paradigm where cloud contamination serves as the sole forward degradation source. This allows the model to precisely simulate the morphology and dynamic evolution of real-world cloud contamination, thereby enhancing network exposure to diverse cloud-affected patterns and continuously varying intensities in the training stage. During inference, an enhanced sampler iteratively applies the restoration network and the cloud-deterministic diffusion operator, progressively reversing the cloud degradation while effectively preserving residual details. Leveraging dynamic cloud-driven transitions and iterative updates, CDPCR robustly tackles heterogeneous real-world cloud contamination and achieves more realistic and efficient sequence-to-sequence (S2S) reconstruction for optical time series. Crucially, it operates without requiring external cloud masks while alleviating the strong dependence on clean temporal data. Extensive experiments across different datasets validate the superiority and robustness of the proposed method.**

***Index Terms*—Cloud contamination, cloud removal, diffusion model, image reconstruction, iterative sampling.**

I. Introduction

REMOTE sensing imagery (RSI) is crucial for a wide range of applications like environmental monitoring [1], [2], natural resource management [3], [4], and disaster response [5], [6]. However, optical RSIs are inherently vulnerable to atmospheric conditions, with cloud contamination being the most common source of degradation. U.S. Geological Survey reports the global average cloud cover reaches about 66%, rendering 30%–40% of RSIs unsuitable for direct applications [7]. Moreover, natural clouds exhibit irregular morphology and high variability in shape, thickness, coverage, and distribution, while the underlying ground features are diverse and subject to temporal changes [8], [9]. Simple strategies, such as discarding cloud-affected images, are inadequate when confronting extensive clouds, particularly in scenarios demanding high spatial accuracy and temporal resolution. Therefore, cloud removal is a crucial preprocessing step for satellite image applications and has emerged as an independent research direction with various specialized approaches.

Previous research typically classifies clouds into thin and thick categories, processing them separately [10], [11]. Thin clouds cause semi-transparent blurring, with valuable information retained in the observations [9]. Accordingly, high-fidelity reconstruction requires fully leveraging this retained information in the spatial and spectral dimensions [10]. Thick clouds are often opaque, occluding information across all bands [12]. Auxiliary data from adjacent temporal images or other sources have become critical information for reconstruction [11], [13]. However, the conventional "thin/thick" cloud classification remains subjective. The boundary between these two types of clouds is ambiguous and lacks standardized quantitative criteria to define them explicitly and scientifically. Furthermore, this coarse dichotomy is too simple to encompass real-world scenes with intermingled cloud layers of varying thickness, especially neglecting heterogeneous clouds with intermediate or mixed states. Models adhering to this classification are insufficient to handle complex cloud contamination and varying degrees of retained information across the cloud-affected regions [8], [9], leading to limited robustness and generalization in natural scenes. Consequently, developing methods to comprehensively address complex cloud contamination across varying thickness and coverage is critical, representing the core challenge in current cloud removal research.

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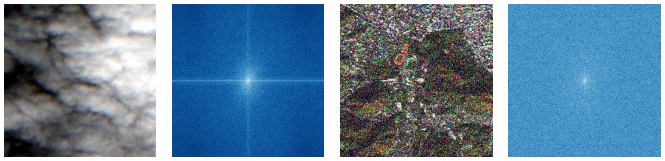
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(a) (b)

Fig. 1. Spectrogram analysis. (a) Cloud contamination versus (b) Gaussian noise.

With the rapid development of deep learning (DL) techniques, numerous studies have employed deep neural networks for cloud removal [14], [15], [16], [17]. These approaches leverage powerful feature representation capabilities, together with large-data training strategies, to address severe and complex cloud contamination and enhance both the quality and efficiency of restoration. Yet they still suffer from the lack of ground truth (GT), model instability, reduced diversity, and limited generalization performance due to inherent limitations of both the model architecture and the cloud removal task. To overcome these bottlenecks, diffusion models have emerged as a promising paradigm due to their self-supervised nature and iterative sampling process [18], [19]. These models are efficient and stable, utilizing a progressive sampling process to ensure the authenticity and diversity of restoration results [20]. Given these characteristics, diffusion models have attracted growing attention for cloud removal, with several studies exploring their potential in this direction [21], [22], [23], [24].

However, existing studies on diffusion-based cloud removal methods generally focus on conditional generation, leveraging long temporal sequences or SAR images as conditions to guide the reverse sampling process [21], [23]. This paradigm requires elaborate data preparation, including cross-source alignment and fusion, and often leads to high computational complexity throughout different stages of the workflow. Moreover, most existing works adopt Gaussian noise diffusion as the forward process [22], [25], which fails to account for the intrinsic complexity and variability of natural cloud contamination, neglecting the physical evolution of cloud degradation. Specifically, we observe a discrepancy in the frequency characteristics between cloud interference and Gaussian noise (see Fig. 1). This fundamental mismatch limits the diversity and representativeness of degraded samples available for training, leaving restoration operators ill-equipped to handle real-world atmospheric conditions. Therefore, instead of relying on stochastic noise, establishing a physically consistent diffusion paradigm that directly aligns with the cloud degradation manifold is imperative for more robust and high-fidelity restoration.

In this study, we propose a cloud-deterministic diffusion and progressive cloud removal (CDPCR) model. To overcome the dual bottlenecks of the binary thin/thick classification and the physical mismatch in standard diffusion paradigms, CDPCR establishes a physically aligned diffusion framework to more effectively handle complex natural cloud contamination through a realistic degradation manifold. Specifically, it is designed with a novel cloud-deterministic forward process to simulate the dynamic evolution of natural cloud contamination and a deterministic backward process for progressive reconstruction from severe to mild states. Under this paradigm, CDPCR enables efficient sequence-to-sequence (S2S) recovery for optical time series, providing a robust and physically aligned solution for real-world cloud contamination. The key features of CDPCR are outlined as follows.

1) *Novel Forward Process:* We introduce the cloud-deterministic diffusion process, which develops a deterministic cloud degradation schedule that utilizes cloudy information as the sole degradation source to ensure realistic data augmentation. This process captures the inherent heterogeneity and variability of natural cloud contamination by simulating its morphology and dynamic progression from simple to complex states, thereby enhancing training exposure to diverse patterns featuring continuously varying cloud thickness and coverage, and significantly boosting generalization against complex real-world cloud contamination.
2) *Robust Sampling Strategy:* We develop a deterministic backward process with an improved iterative sampling algorithm. By alternately applying the restoration network and the cloud-deterministic diffusion operator, the sampler reverses the cloud degradation while effectively preserving interstep residual details during inference. This deterministic sampling ensures an intuitive reconstruction path from severely to mildly contaminated states, progressively enhancing image fidelity and diversity in the final output.
3) *Flexibility and Independence:* CDPCR operates in a self-supervised manner, eliminating dependencies on GT. Furthermore, its specialized cloud-deterministic diffusion process and iterative learning strategy collectively remove reliance on external data sources, including cloud and shadow masks. Moreover, the framework demonstrates high structural adaptability to varying spectral bands and temporal lengths, enabling efficient S2S recovery for optical time series across different data conditions. Extensive experiments validate the superiority and robustness of the proposed method.

The remainder of this article is organized as follows. Section II reviews related work. Section III details the proposed CDPCR framework. Sections IV and V present the experimental comparisons and ablation studies, respectively. Finally, Section VI concludes this article.

II. Related Works

In this section, we first review existing cloud removal methods. Next, we introduce the fundamental principles underlying diffusion models, followed by a brief overview of recent diffusion-related variants and applications.

A. Cloud Removal

Existing studies generally employ two main approaches to address cloud contamination [10], [11]. In the early stage when technologies and data are limited, single-image-based cloud removal methods are primarily employed [26]. These methods fully excavate available spatial and spectral information retained in the cloud-contaminated imagery for

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reconstruction and play an important role in handling milder cloud contamination, where clouds are relatively thin, small, and sparse [9], [27], [28], [29]. In scenes with complex (e.g., varying thickness) and intense cloud contamination, where spatial and spectral information are severely degraded, external information is crucial for reconstructions. Therefore, multi-image-based approaches are developed for cloud removal in RSIs, including multitemporal [12], [13] and multisource methods [30], [31], [32]. Traditionally, these methods utilize linear correlations among data [33], [34], [35] or related physical and mathematical priors to reconstruct missing regions [11], [36], [37]. Linear methods are easy to implement but yield uncertain results, heavily dependent on observation conditions. Physics-based methods produce more refined cloud removals, yet involve intricately hand-crafted priors, complicated solving processes, and manual parameter tuning, which risk introducing uncertainties and reduced discriminability in results, and severely limiting flexibility and efficiency in large-scale applications [13], [38]. Furthermore, these traditional models struggle with extensive and persistent thick clouds, and the quality of their results is susceptible to the accuracy of external cloud/shadow masks [11], [36]. Therefore, to more effectively address complex and severe cloud contamination, DL has been widely adopted for the cloud removal task.

With the development of DL techniques, various architectures are utilized for cloud detection and removal, including convolutional neural network (CNN)-based methods [39], [40], [41], generative adversarial network (GAN)-based approaches [15], [42], [43], and others. Advanced approaches like Transformer-based models [44], [45], [46] and data fusion strategies are further explored for better processing, exemplified by the lightweight CloudViT [47] for cloud detection and HS2P [48] which ensures spectral and structural fidelity of reconstructions through hierarchical fusion, proving the potential of DL in complex scenes. By harnessing powerful feature representation capability and large-data training strategies, DL techniques offer an efficient and effective solution for severe cloud contamination [8], [49], [50]. However, the absence of GT is a fundamental challenge for DL-based cloud removal in optical RSIs, hindering supervised training and the quantitative evaluation of these deep models. To overcome this, some methods construct synthetic cloudy/clean image pairs to enable supervised learning. Yet simulated clouds inevitably differ from real ones, restricting the generalization of trained deep models in real-world scenarios. Other strategies utilize adjacent clean temporal observations as GT proxies. Nevertheless, land-cover changes introduce temporal discrepancies that prevent strict correspondence between paired images, thereby undermining the reliability of restoration results. This issue remains a major bottleneck for such approaches. Moreover, DL-based direct mapping neglects the intrinsic properties of RSIs and cloud contamination, leading to degraded image fidelity and unstable generalization performance in real-world scenes [40]. Although deep internal learning introduces deep priors for unsupervised learning of image properties [49], [51], it faces challenges in determining the optimal stopping condition and requires further regularization [52], [53]. GAN-based methods [54], [55], conversely, suffer from mode collapse, impairing data quality and diversity. Transformer- and attention-based models [15], [16] often incur high computational costs and require complex data preparation processes. Collectively, these inherent limitations underscore the need for a more robust, stable, and nonlinear solution.

Recently, diffusion models have emerged as a promising paradigm for vision tasks [19], [56]. Utilizing dynamic and iterative transformations, they generate repaired images from self-degraded ones, laying the foundation for self-supervised learning [20]. This mechanism enables iterative DL and adaptive image modeling, ensuring intuitive alignment with the progressive reconstruction process from severe to mild [57]. Consequently, these inherent capabilities make diffusion models well-suited to address inherent challenges of the cloud removal task (e.g., strong dependence on GT and difficulty in removing complex contamination). Increasingly, diffusion models have become an active research direction for cloud removal. These studies primarily focus on conditional generation, which leverage cloudy sequences or stronger auxiliary data like SAR images, as conditions to guide reverse sampling and reconstruction [21], [23], [24]. This approach often involves elaborate and time-consuming data preparation, including cross-source alignment and fusion, leading to high computational complexity. Moreover, existing studies generally adopt the forward process with Gaussian noise injection [22], [23], [25]. However, this paradigm neglects the real degradation process in cloud-affected RSIs, failing to fully capture the inherent complexity and variability of real-world cloud contamination, nor does it align with the dynamic progression of natural contamination. This fundamental mismatch results in unstable generalization performance of the methods. Therefore, the degradation source in the forward process warrants further exploration, aiming for a more suitable data augmentation strategy for cloud removal model training. Recently, degradation-guided, cloud-simulation-based, and diffusion bridge models, such as EDM-CR [21] and DB-CR [58], have advanced cloud removal by incorporating task-specific forward trajectories and multimodal SAR guidance. However, these frameworks still implicitly rely on Gaussian noise or Brownian perturbations to govern their forward evolution, thereby introducing stochastic fluctuations into the degradation trajectories. While effectively leveraging cloud degradation priors or SAR structural guidance under severe occlusions, these approaches rely heavily on paired multimodal auxiliary inputs and precise multisensor alignment.

To this end, we propose a novel noise-free diffusion paradigm, termed cloud-deterministic diffusion. Unlike standard Gaussian noise diffusion, our approach employs natural cloud contamination as the sole forward degradation type. This allows the model to better capture the morphology of real-world contamination and simulate its dynamic evolution (from simple to complex states), thereby offering an effective and robust solution for complex and severe cloud contamination in RSIs. Moreover, CDPCR establishes an efficient S2S reconstruction framework. In contrast to sequence-to-point (S2P) recovery [23], [42], CDPCR produces complete reconstructions of the observed sequences, thereby retaining richer temporal information and meeting the requirements for

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long time series and high temporal resolution [4], [59] within a flexible and unified architecture.

B. Diffusion Models

Inspired by thermodynamics, diffusion can be understood as a random walk beginning from a random state with heavy noise and slowly annealing into a smooth state with little noise [18]. Diffusion models feature a forward diffusion with gradual noise addition and a backward inference with progressive noise removal [20]. Specifically, the model comprises two core components: the noise diffusion operator $\mathcal{D}$ for degradation and the trained restoration operator $\mathcal{R}$ for sampling and generation. By alternately applying these two operators, diffusion models present a complete generative process, comprehensively learning the underlying distributions of training data and progressively producing high-quality images with rich sample diversity. Random Gaussian noise diffusion is common and standard from the community understanding of the diffusion literature [56], [60]. The general paradigm of the forward process is as follows:

$$\mathbf{X}_s = \mathcal{D}(\mathbf{X}_0, s), \quad s \in \{S, S-1, \ldots, 1, 0\} \tag{1}$$

$$\mathcal{D}(\mathbf{X}_0, 0) = \mathbf{X}_0, \quad \mathcal{D}(\mathbf{X}_0, S) = \mathbf{X}_S \tag{2}$$

where $\mathbf{X}_0$ is the original clean image. The variable $\mathbf{X}_s$ denotes the degraded representation of $\mathbf{X}_0$ at step s. S defines the total number of diffusion steps. Then, a restoration network ($\mathcal{R}$) is trained to predict the clean image $\mathbf{X}_0$ from $\mathbf{X}_s$ at different steps. During the backward process, a sampling rule is applied to iteratively update the restoration results, progressively reversing the noise diffusion process and producing the final image [18]. Notably, the sampling methods of diffusion models can be categorized into two types—stochastic [60] and deterministic [61]—depending on whether random Gaussian noise is introduced during the backward process

$$\mathcal{R}(\mathbf{X}_s, s) = \widehat{\mathbf{X}}_0 \approx \mathbf{X}_0. \tag{3}$$

As is known, diffusion models come in many flavors [20], [62], [63]. Earlier studies on diffusion models have incorporated experiments involving deterministic image generations [61], [64] and selected inverse problems [56]. Recently, diffusion models have been further applied to tackle issues of deblurring, denoising, super-resolution, and compressive sensing [62], [65] and have emerged as an effective and stable tool for generative modeling [19]. Recently, the role of Gaussian noise in diffusion models has been questioned [66], [67]. Through these theoretical and empirical discussions, diffusions that require no Gaussian noise during training and testing were introduced [66]. Unlike standard diffusion, they refer to the forward processes with other deterministic and smooth deformations like blurring, pixel masking, snowification, and animorphosis [56], [62]. In these forward processes, the degradation routines are constant, and the final distribution is frozen and predictable. It indicates the potential of more generalized diffusion processes with arbitrary transformations for different restoration tasks.

III. Methodology

A. Overall Framework of CDPCR

In the cloud removal task, the observed cloud-contaminated multitemporal RSIs can be represented as a fourth-order tensor $\mathbf{Y} \in \mathbb{R}^{H\times W\times C\times T}$, where T denotes the number of temporal frames, and C, H, and W represent the channels, height, and width, respectively. In general, $\mathbf{Y}$ consists of foreground cloud layers $\mathbf{C} \in \mathbb{R}^{H\times W\times C\times T}$ and underlying images $\mathbf{X} \in \mathbb{R}^{H\times W\times C\times T}$ [36]. $\mathbf{C}$ and $\mathbf{X}$ have distinct characteristics and interact with each other, while the clouds themselves also exhibit varying properties [35]. The complexity and variability of cloud-contaminated RSIs underscore the necessity of specific cloud learning for more realistic and robust cloud removal.

In this study, we propose the cloud-deterministic diffusion model (CDPCR), a novel cloud removal framework for progressive reconstruction of cloud-contaminated RSIs. The goal of CDPCR is to offer an effective and robust solution for complex and severe cloud contamination in real-world scenes, providing complete temporal processing of cloud-affected data with arbitrary temporal lengths and spectral bands, across varying severity. Specifically, our model is based on treating cloud contamination as a deterministic diffusion process and assumes it is the only degradation present. It models a forward process that progressively degrades the clean background with natural cloud contamination, where the final diffused state approximates the observations. This cloud-deterministic diffusion serves as an inherent data augmentation mechanism, incorporating varying forms and degrees of cloud interference during training. Leveraging this forward process, we train an image restoration network for real-world cloud removal, enabling it to effectively handle diverse cloud thickness, coverage, and distribution. Subsequently, we define a deterministic backward process equipped with an improved sampling algorithm to iteratively recover the clean image from any given diffused state. By alternating between the restoration network and the degradation operator, CDPCR progressively achieves thorough cloud removal and realistic ground feature reconstruction. Fig. 2 illustrates the overall framework of the proposed model.

During the training stage, the model is trained on paired samples, each consisting of a T-frame cloudy sequence for degradation and a single clean temporal image (<5% cloud cover) as the corresponding underlying background. At inference time, the model exclusively receives T-frame cloudy image sequences to perform real-world cloud removal and yields complete temporal processing (S2S recovery) of the contaminated data. Crucially, the model can flexibly adjust both the temporal length (T) and the number of spectral channels to meet varying application demands and data availability in practical scenarios. Section V further validates the performance of CDPCR with different temporal and spectral input settings. Moreover, leveraging self-supervised iterative learning and the specialized cloud-deterministic diffusion process, CDPCR fundamentally reduces dependence on GT and external data sources, including cloud and shadow masks. By efficiently capturing the morphology and dynamic evolution

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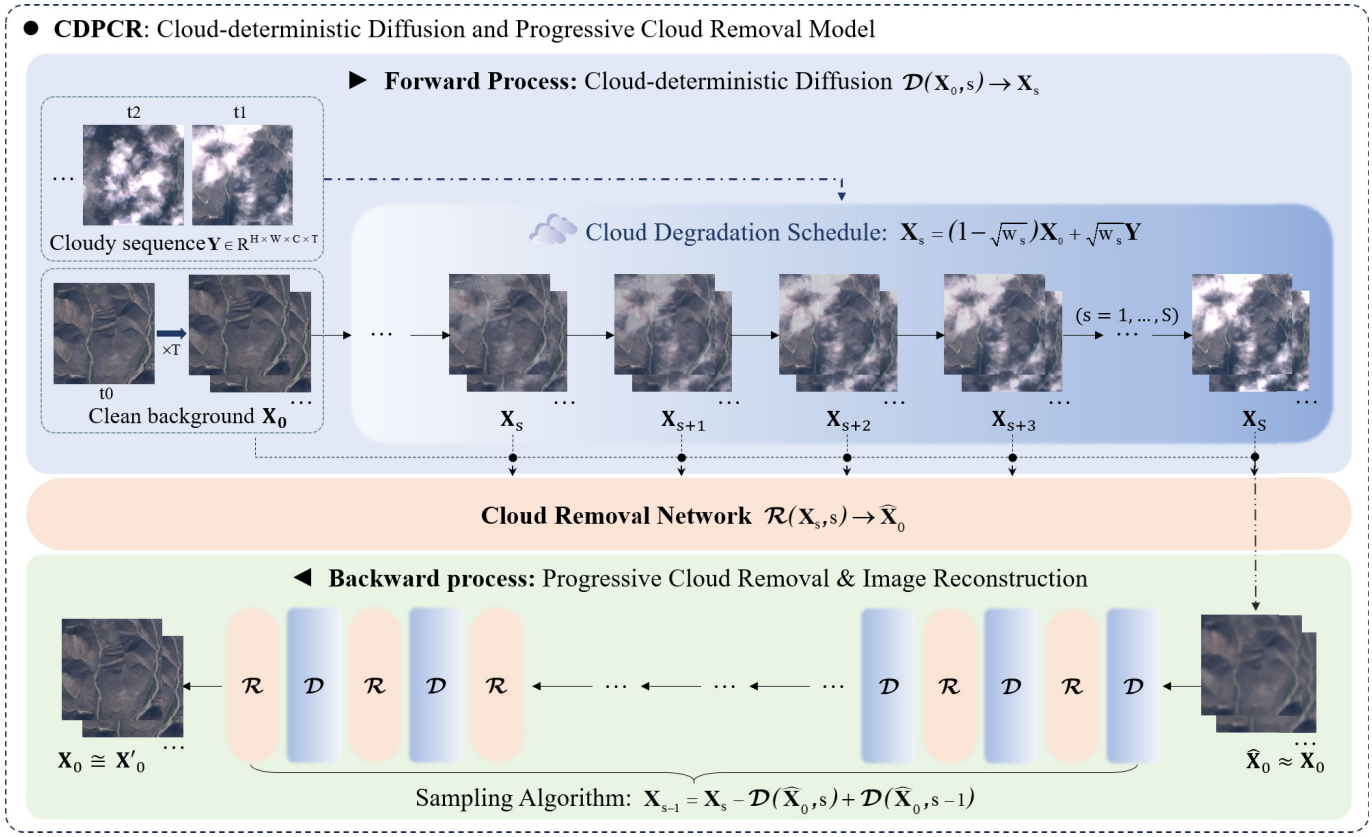


Fig. 2. Overall framework of the proposed CDPCR. The model consists of a forward process that degrades the clean background with varying cloud contamination and a reverse process that removes it, progressively reconstructing the background of the cloud-affected sequence.

(from mild to severe states) of natural cloud contamination, this novel design also avoids the unnatural cloud simulation process and the reliance on adjacent clean temporal observations as GT proxies. Overall, CDPCR introduces a new diffusion framework based on real-world cloud degradation. It fully considers the unique properties of natural cloud contamination by incorporating them into the forward degradation process, which enables it to leverage the inherent diversity, complexity, and variability of natural clouds for model optimization. By treating cloud removal as a continuous reconstruction task along a degradation manifold, CDPCR moves beyond the binary thin/thick dichotomy, adaptively handling intermingled cloud layers of varying thickness. Consequently, it effectively addresses diverse and complex cloud contamination in real-world scenarios, achieving robust and realistic recovery of observed optical RSIs. The technical details of CDPCR are presented in Sections III-B–III-D.

B. Forward and Backward Processes

Natural cloud contamination and Gaussian noise exhibit distinct characteristics (see Fig. 1). This fundamental mismatch limits the range and type of augmented data available for training of diffusion models. As a result, restoration operators trained on Gaussian noise are ill-equipped to handle real-world cloud degradation in RSIs, underscoring the necessity of re-examining the selection and impact of the degradation source in the forward process. This raises two pivotal questions: can real-world cloud contamination in RSIs serve as the forward degradation source to enhance cloud removal effectiveness? and how can we develop a degradation schedule that ensures the efficient optimization of this task?

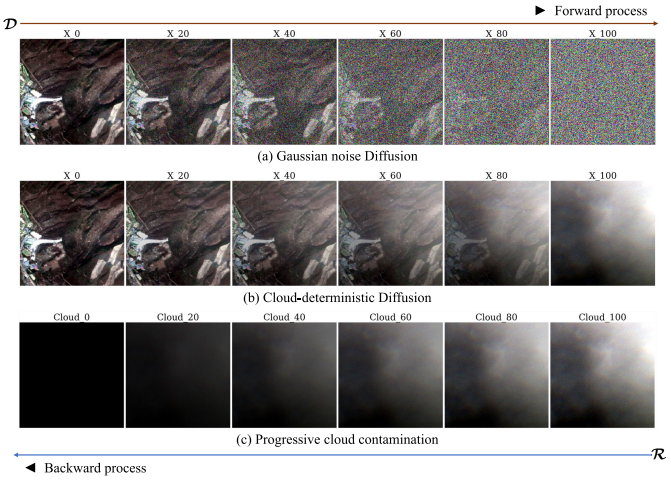


Fig. 3. Schematic of diffusion processes. (a) Standard Gaussian diffusion process versus (b) proposed cloud-deterministic diffusion process. (c) Visualizations of natural cloud contamination progressively intensified during the forward process. Note that $\mathbf{X}_0$ denotes the clean background, while $\mathbf{X}_{100}$ represents the final degraded state.

To overcome it, CDPCR introduces a new diffusion paradigm to effectively target natural cloud contamination in RSIs. Unlike standard Gaussian noise diffusion [see Fig. 3(a)],

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it establishes a more intuitive forward process explicitly modeling real-world cloud degradation, accomplishing dynamic progressions from cloudless backgrounds to naturally cloudy images [see Fig. 3(b)]. The forward and backward processes of CDPCR are detailed in Subparts 1) and 2).

1) Forward [Cloud-Deterministic Diffusion Process ($\mathcal{D}$)]: Forward diffusion acts as a data augmentation step, providing the restoration network with more diversified distributions to enhance training stability and improve performance in low-density regions [60], [63]. Considering the inherent variability and complexity of natural clouds, we design a novel cloud-deterministic diffusion process accomplished by a linear-weighted degradation schedule, formulated as (4). This schedule progressively diffuses natural cloud contamination from observed cloudy sequences to background clean images, enabling smooth transitions between cloud-free and cloud-affected states with varying levels of contamination. Therefore, it allows the model to be trained on augmented datasets with diverse, continuously varying cloud thickness and coverage

$$\mathbf{X}_s = \left(1 - \sqrt{w_s}\right)\mathbf{X}_0 + \sqrt{w_s}\mathbf{Y}. \tag{4}$$

Specifically, we develop a cloud-deterministic diffusion operator $\mathcal{D}$ effecting a transition from cloudless to cloudy, aiming to model cloud contamination in the real world. As the core of the forward process, $\mathcal{D}$ defines a closed-form, deterministic routine via (4) to blend the clean background X_0 with the cloudy observation Y. This routine is governed by a deterministic degradation schedule, where the intensity is regulated by a sequence of linearly spaced weights, denoted as $\{w_s\}_{s=0}^{S}$. The w_s controls the varying degrees of coverage and thickness of the cloud contamination imposed on the clean background $\mathbf{X}_0$ (representing the underlying image $\mathbf{X}$) during the forward process. To construct a continuous degradation trajectory, w_s is modeled as a linear function of the timestep s, which can be explicitly formulated as: $w_s = s/S$, where S denotes the total number of forward diffusion steps. As s advances from 0 to S, w_s strictly scales from 0 to 1.

During the cloud-deterministic diffusion process, the operator $\mathcal{D}$ generates the image state $\mathbf{X}_s$ at each step s as a weighted blend of the clean background $\mathbf{X}_0$ and the cloud contamination. As w_s varies from 0 to 1, the resulting sequence $\{\mathbf{X}_s\}_{s=0}^{S}$ comprises images with diverse and progressively intensifying cloud coverage and thickness, where the length of $\{\mathbf{X}_s\}_{s=0}^{S}$ depends on the number of diffusion steps S. The formulation of $\mathbf{X}_s$ at each diffusion step s is expressed in (4), where w_s controls the intensity of cloud contamination from the observation $\mathbf{Y}$ blended into the background $\mathbf{X}_0$ at step s.

Unlike the stochastic noise injection in standard diffusion models, the proposed cloud-deterministic forward process introduces no random noise. In this diffusion framework, $\mathbf{Y}$ is nonstochastic, serving as the degradation source and the terminal state of the forward pass (i.e., $\mathbf{X}_S$). Following (4), the cloud-deterministic diffuser $\mathcal{D}$ progressively transforms the clean background images into cloudy states through a series of diffusion steps. This progressive transformation enables continuous transitions from cloud-free to cloudy conditions with varying contamination levels, thereby facilitating comprehensive representation learning across cloud-affected regions.

In this paradigm, the proposed cloud-deterministic diffusion trajectory implicitly captures the dynamic evolution of clouds, mirroring the continuous variation in cloud morphology and intensity across multitemporal cloud-affected sequences $\mathbf{Y}$. This design allows the model to learn the intrinsic degradation manifold of natural cloud contamination, which is crucial for effective restoration. During the inference phase, the naturally cloudy sequence $\mathbf{Y}$ directly serves as both the observation and the starting point of the restoration process.

Crucially, the cloud-deterministic diffusion process provides a novel strategy for cloud degradation modeling, which narrows the domain gap between modeled and observed cloud distributions. Unlike standard Gaussian noise, this cloud-driven diffusion approach functions as an implicit data-driven physical prior. By modeling the degradation manifold based on actual atmospheric conditions, the network is constrained to learn the specific spectral and spatial characteristics of cloud interference rather than random noise. This ensures that the model is optimized within a realistic physical context, effectively narrowing the domain gap and mitigating the generation of unphysical textures in the reconstruction stage.

Furthermore, the heterogeneity is technically introduced through two synergistic dimensions in CDPCR: the internal diffusion trajectory, which models the progression of intensities, and the external data diversity, where the inherent spatial heterogeneity of natural samples ($\mathbf{Y}$) provides the structural basis for $\mathcal{D}$ to simulate realistic cloud patterns. Specifically, the contamination imposed by the diffusion operator $\mathcal{D}$ encompasses a wide spectrum of atmospheric degradation effects. $\mathcal{D}$ introduces heterogeneous cloud morphologies, ranging from translucent thin mist and scattered broken clouds to dense, opaque occlusions. Moreover, it accounts for the optical interactions associated with cloud cover, such as the casting of realistic cloud shadows and the reduction of ground visibility proportional to cloud thickness. By progressively intensifying these complex features during the forward process, the model is exposed to a continuous and realistic degradation manifold, enabling it to learn robust restoration capabilities across diverse cloud scenarios, thereby improving the model generalization performance in real-world scenes. Fig. 3(b) and (c) visually demonstrates the progressive visual evolution during this cloud-deterministic diffusion process.

2) Backward [Progressive Cloud Removal and Image Reconstruction ($\mathcal{R} \leftrightarrows \mathcal{D}$)]: Natural cloud contamination exhibits inherent heterogeneity, where underlying details often remain discernible under thin or sparse clouds. To capture and embed these varying visibility states into the learning process, the forward process gradually blends observed cloudy patterns into clean background images. Consequently, the final diffused distributions approximate real-world observations, providing the necessary physical priors for the backward phase. The backward process is designed to reverse this cloud-deterministic degradation, underscoring the critical role of information preservation in the improved sampling method. Through iterations, the sampler progressively achieves thorough cloud removal and image reconstruction with realistic ground details. Figs. 2 and 4 illustrate this collaborative sampling process.

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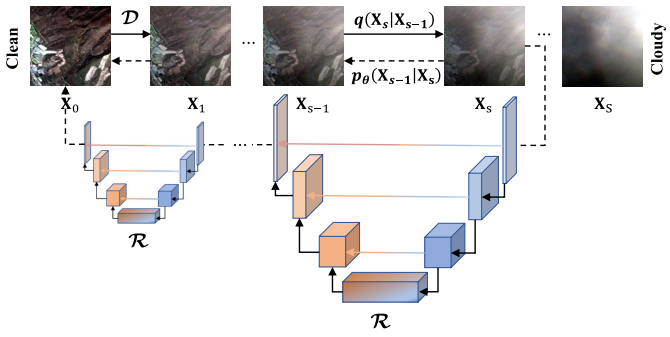


Fig. 4. Collaborative backward sampling mechanism in CDPCR. The restoration network $\mathcal{R}$ and diffusion operator $\mathcal{D}$ alternate between image correction and cloud reprojection, iteratively reversing the degradation to refine details.

Specifically, the backward process employs an image restoration network $\mathcal{R}$ and the cloud-deterministic diffusion operator $\mathcal{D}$ to iteratively invert the cloud contamination. The restoration network $\mathcal{R}$ serves as the core cloud removal engine; its detailed mathematical architecture, tensor formats, and block designs are elaborated in Section III-C and illustrated in Fig. 4. During the generative process, the sampler alternates between $\mathcal{R}$ for image correction and $\mathcal{D}$ for enforcing the degradation constraint, using an improved sampling strategy tailored to deterministic and smooth degradation [66]. As outlined in Algorithm 2, this iterative mechanism progressively refines the reconstructed images. Compared with the standard sampler (detailed in Algorithm 1 in Appendix A), this improved algorithm exhibits higher robustness against potential restoration errors, particularly for small values of s. In particular, the residual term in the improved update rule (Algorithm 2), $\mathbf{X}_s - \mathcal{D}(\widehat{\mathbf{X}}_0, s)$, is designed to preserve more high-frequency information during inference. This term stabilizes the deterministic sampling process for cloud removal, as it compensates for potential reconstruction inaccuracies by maintaining the underlying clean image structure from the previous estimation. Consequently, this process ensures an intuitive reconstruction path from severely to mildly contaminated states, progressively enhancing image fidelity and diversity in the final output.

To further clarify the implementation of the alternating $\mathcal{R} \leftrightarrows \mathcal{D}$ sampling procedure, the degradation operator $\mathcal{D}$ is used differently during training and inference. During training, $\mathcal{D}$ acts as a closed-form degradation synthesizer, directly generating the corrupted state $\mathbf{X}_s$ for an arbitrary randomly sampled timestep s in a batch-parallel manner. This avoids explicitly traversing intermediate states and enables the restoration network $\mathcal{R}$ to learn clean-background estimation under different cloud-contamination levels.

During reverse sampling, in contrast, $\mathcal{D}$ is incorporated into the iterative state transition as a deterministic residual correction. Given the estimated clean background $\widehat{\mathbf{X}}_0$ at step s, the transition is implemented as

$$\mathbf{X}_{s-1} = \mathbf{X}_s - \left(\sqrt{w_s} - \sqrt{w_{s-1}}\right)\left(\mathbf{Y} - \widehat{\mathbf{X}}_0\right). \tag{5}$$

Thus, the training-time operation provides efficient supervision across arbitrary degradation levels, whereas the inference-time operation progressively removes cloud artifacts along the deterministic restoration trajectory. This implementation also avoids redundant forward degradation steps and helps to preserve high-frequency structural information during reconstruction.

C. Image Restoration Network $\mathcal{R}$

As illustrated in Fig. 4, the restoration network $\mathcal{R}$ adopts a timestep-conditioned U-Net. Given a degraded state $\mathbf{X}_s \in \mathbb{R}^{B\times C_{\text{in}}\times H\times W}$ at restoration step s, it produces $\mathcal{R}(\mathbf{X}_s, s) \in \mathbb{R}^{B\times C_{\text{out}}\times H\times W}$, which is used to estimate the clean image $\hat{\mathbf{X}}_0$. For multitemporal RGB observations concatenated along the channel dimension, $C_{\text{in}} = C_{\text{out}} = 9$.

The backbone follows a symmetric four-scale encoder–decoder hierarchy with feature widths $(C, 2C, 4C, 8C)$, where C is the base channel dimension. An initial convolution projects the input to C channels. At each encoder scale, two ConvNeXt blocks extract localized features before a 4×4 stride-2 convolution performs downsampling. A central bottleneck bridge comprising two ConvNeXt blocks and a linear attention module processes the deepest feature representations at resolution $8C$. The decoder mirrors the encoder structure, utilizing 4×4 stride-2 transposed convolutions for upsampling. Encoder features are concatenated with the corresponding decoder features through long skip connections, preserving fine spatial details. A final convolution maps the decoded features to C_{out} channels.

Each ConvNeXt block consists of a 7×7 depthwise convolution, channelwise LayerNorm, a 1×1 expansion convolution, GELU activation, and a 1×1 projection convolution. Residual connections use identity mappings when the input and output dimensions match, and 1×1 convolutional projections otherwise. The hidden channel width is expanded by a factor of 2. The scalar timestep s is encoded by sinusoidal positional embeddings and refined by a two-layer MLP to obtain a temporal embedding $\mathbf{t} \in \mathbb{R}^C$. This embedding is broadcast spatially and added to the intermediate features of every ConvNeXt block after depthwise convolution

$$\text{Linear}(C, 4C) \rightarrow \text{GELU} \rightarrow \text{Linear}(4C, C).$$

To model long-range spatial contexts efficiently, a linear attention module with residual prenormalization (Residual(PreNorm(·))) is deployed at each scale of the encoder, bottleneck, and decoder. By normalizing keys along the spatial dimension via Softmax, the global attention context is computed with linear computational complexity $\mathcal{O}(N)$ relative to the number of spatial locations N. Together, local ConvNeXt feature processing, temporal conditioning, and efficient global attention enable $\mathcal{R}$ to progressively reconstruct cloud-free multitemporal observations during reverse restoration.

D. Objective Function and Optimization

The optimization of CDPCR is designed to ensure robust restoration across complex cloud contamination. Specifically, the proposed cloud-deterministic forward diffusion exposes the model to diverse cloud patterns and intensities, facilitating effective optimization and robust generalization. During

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training, the restoration network $\mathcal{R}$ is equipped with a time embedding strategy [60] to estimate the underlying clean image from inputs at different diffusion timesteps. The network parameters θ are optimized to recover the original clean background $\mathbf{X}_0$ by minimizing the reconstruction loss $\mathcal{L}(\theta)$. Formally, we primarily employ an L_1 loss function to measure the difference between the reconstructed image $\widehat{\mathbf{X}}_0$ and the GT $\mathbf{X}_0$. Under this optimization scheme, the restoration network is robustly trained to capture the necessary clean image information

$$\mathcal{L}(\theta) = \mathbb{E}\left[\left\|\mathbf{X}_0 - \widehat{\mathbf{X}}_0\right\|_1\right]. \tag{6}$$

At the inference stage, CDPCR executes a collaborative backward process driven by the synergy between the cloud removal network $\mathcal{R}$ and the cloud-deterministic diffuser $\mathcal{D}$. These components work in tandem to iteratively reverse the cloud-deterministic degradation across all timesteps, progressively refining the image content to recover clean background details. During testing, the model is configured to exclusively accept T-frame cloudy sequences as input. Notably, the framework is designed to adaptively handle variable-length input/output sequences, allowing the application of a unified model to diverse needs and data conditions. In real-world experiments, this framework is directly applied to observed contaminated data, ensuring reliable reconstruction in natural scenes. For quantitative evaluation, the process operates on synthetic inputs generated by $\mathcal{D}$ from clean backgrounds ($\mathbf{X}_0$), ensuring a consistent contamination standard with explicitly available GT ($\mathbf{X}_0$) for unambiguous metric calculation. In summary, the proposed CDPCR framework offers significant operational advantages. Unlike conventional simulation-based models, CDPCR directly operates on naturally cloudy RSIs during both training and testing. This effectively bridges the domain gap, thereby ensuring robust generalization capability in real-world scenarios. By leveraging dynamic cloud-aware transitions and iterative learning, CDPCR also eliminates the need for external cloud masks and minimizes reliance on clean temporal References, enabling efficient S2S recovery.

IV. Experiments and Results

A. Dataset Preparation and Experimental Settings

To comprehensively evaluate the performance of CDPCR, we conduct experiments on multiple datasets. These datasets, featuring various ground features and complex cloud contamination (including diverse distribution, coverage, morphology, and thickness), provide ample opportunity for the model to learn the inherent heterogeneity and variability of real-world cloud contamination in optical RSIs.

1) Datasets and Input Configuration:

a) SEN12MS-CR-TS: We utilize the SEN12MS-CR-TS dataset [50], a large-scale public dataset for multitemporal cloud removal, providing observations with regular revisits across global, all-season coverage. The dataset contains 30-frame temporal series of Sentinel-2 (S-2) imagery, consisting of 256×256 13-band imagery patches. For training, we utilize three-frame cloudy S-2 sequences (with dates t_1–t_3) as degraded observations, each paired with a cloudless S-2 image selected as the clean background $\mathbf{X}_0$. The date of $\mathbf{X}_0$ (t_0) is carefully chosen to fall within the temporal range of the observed sequence, ensuring temporal proximity and consistency (e.g., resulting in an order like t_1, t_0, t_2, and t_3). During testing, only the three-frame cloudy sequences are input for real-world reconstruction. For simulated evaluation, we synthesize cloudy inputs via the cloud-deterministic diffusion process, blending contamination onto the clean background for iterative recovery.

Furthermore, to analyze the stability and performance fluctuation of CDPCR, we evaluate the model using image sequences with complex cloud contamination featuring varying patterns and intensities. To facilitate explicit and systematic evaluation, we categorize the observations into different types, based on the severity of cloud interference and the quantity of available cloudless information (see Fig. S1 in the Supplementary Material). Specifically, these tags define the temporal cloud status of each frame: "0" signifies clean images, "1" denotes images affected by clouds/shadows (e.g., "S010" is cloudless-cloudy-cloudless), and "2" indicates cloud-affected images over highly reflective surfaces (e.g., snow). Overall, this system yields six distinct observation types (S010-S111), with the S110 type further subdivided into S110-a and S110-b to account for distinctions in surface features. Contamination severity increases as the number of cloudy images ("1") rises. This systematic categorization ensures a comprehensive evaluation of the model performance in real-world scenarios, leveraging the full range of contamination variance (Figs. S1 and A1).

b) Sen2_MTC: We further evaluate CDPCR on the Sen2-MTC dataset [14]. This dataset consists of 50 nonoverlapping tiles (each containing 70 S-2 images with pixel values in the range $[0, 10\,000]$), provided as 256×256 patches with four channels, organized into three-frame cloudy sequences, each paired with a temporally clean image. Unlike the SEN12MS-CR-TS dataset, which emphasizes diversity and variability, the Sen2-MTC dataset provides a critical density contrast: its temporal sequences are consistently affected, with all three frames exhibiting a comparable, significant level of contamination (see Fig. 5). It includes images with frequent, severe cloud contamination and distinct cloud shadow effects, posing a direct challenge for model testing.

For all experiments, the training pair consists of a three-frame cloudy sequence and a corresponding cloudless image ($\mathbf{X}_0$). For stable training, all pixel values in the two datasets are first normalized to the range of $[-1, 1]$. We use mutually exclusive 256×256 patches for model training and testing. For real-world evaluation, we qualitatively assess CDPCR's performance directly on observed naturally cloud-contaminated sequences. Quantitative evaluation is performed through simulated experiments on synthetic inputs generated via the cloud-deterministic diffusion process, allowing us to calculate metrics against unambiguous GT ($\mathbf{X}_0$).

2) Experimental Setup and Evaluation: We optimize the model using Adam [68] with an initial learning rate of $2e^{-5}$ and no weight decay. The model is trained for 100 000 steps with a batch size of 8. An exponential moving average (EMA) [69] is applied to model parameters from the first iteration,

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updating after each step with a decay rate of 0.995 to enhance stability and prevent overfitting. In the forward process, the diffusion time step S is set to 100, and the weight schedule increases linearly from 0 to 1. Experiments are conducted on a Linux platform with an Intel Core i9-9900K CPU and four NVIDIA RTX 3090 GPUs (24 GB each). The operating system used is Ubuntu 18.04, with all networks implemented in Python 3.8 and PyTorch 1.13.0 with CUDA 12.4.

To evaluate CDPCR against existing state-of-the-art (SOTA) methods, we selected nine baseline algorithms, consisting of two physical methods: min filter and mean filter; and seven DL-based methods: UNet, ResNet, STGAN [42], UnCRtainTS [17], PMAA [41], U-TILISE [70], and DiffCR [23]. The DL baselines cover major architectural types: STGAN is used to assess GAN performance; PMAA is a progressive autoencoder; and U-TILISE represents S2S learning models. DiffCR is specifically included as the primary diffusion-based comparison. We also reference UnCRtainTS for its role in establishing the benchmark SEN12MS-CR-TS dataset. All DL-based models were trained using the same datasets and input conditions as CDPCR. Their hyperparameters were slightly tuned based on original paper recommendations to achieve optimal performance. Methods requiring explicit cloud/shadow masks (e.g., U-TILISE) utilized IHOT [71] and S2PixelCloudDetector [72] for mask generation. Notably, we included both S2P and S2S methods in our experiments. Given their difference, we present the single output for S2P methods, while for S2S models, we specifically demonstrate the reconstructed results corresponding to the date with the most severe cloud contamination. Complete S2S reconstructions generated by CDPCR for both datasets are provided in Figs. S3 and S6 of the Supplementary Material.

We adopt five standard indicators to comprehensively evaluate the different methods: mean squared error (mse), mean absolute error (MAE), peak signal-to-noise ratio (PSNR), structural similarity (SSIM) [73], and learned perceptual image patch similarity (LPIPS) [74]. MSE and MAE reflect the quality of spectral restoration. PSNR and SSIM are widely used image quality metrics, where PSNR is based on the signal-to-noise ratio and SSIM measures structural similarity. LPIPS is specifically included to measure perceptual and semantic similarity using a DL-based approach. For interpretation, lower values of mse, MAE, and LPIPS indicate better performance, while higher values of PSNR and SSIM indicate better results. All visualized images are displayed using the RGB mode with 2% linear stretching.

B. Experimental Results on SEN12MS-CR-TS

1) Visualization of Reconstructions Across Varying Cloud Contamination Levels: We present the cloud removal results on the SEN12MS-CR-TS dataset [50] in Figs. 5, 6, and 11. In these figures, clearer structures and finer details compared with the direct network output (CDPCR-R) and CDPCR-S denote the images reconstructed by the restoration network $\mathcal{R}$ and the final output after the sampling process, respectively. While temporally clean images are provided for visual reference, crucially, they are strictly excluded from the model input during inference for real-world cloud removal experiments. The observed sequences showcase diverse surface types (e.g., bare land, forests, urban areas, and waters), affected by clouds of varying shape, thickness, coverage, and distribution. These sequences are diversified due to unpredictable atmospheric conditions and repeated satellite collections in real-world scenes, where the absence and presence of cloudless images within the observations are both common. The difficulty of reconstruction increases as the amount of cloud-free information in the input observation sequence diminishes. To systematically demonstrate model performance and variations, we categorized and arranged these scenarios based on different levels of cloud contamination.

Specifically, these figures present the cloud removal results for different observation types (S010-S111), arranged from mild to severe contamination. In the visualizations, real-world reconstructions are primarily displayed, as CDPCR's forward process is directly modeled upon the characteristics of natural cloud contamination, ensuring that the final diffused states closely resemble real-world cloudy images. For fairness, the compared models conduct the same mapping from natural cloudy to cloudless images. In real-world experiments, restored results are compared with available clean information in the observations; when clean observations are insufficient, temporally clean images (selected for close proximity) serve as the primary reference. For simulated experiments, CDPCR blends the cloud contamination onto the clean background $\mathbf{X}_0$, and then recovers the underlying clean image, where the GT ($\mathbf{X}_0$) is unambiguous.

Fig. 11 presents a comprehensive analysis of model performances across scenarios with varying cloud contamination levels, from mild to severe (S010–S111). When clouds are small and thin [S010, Fig. 11(a)–(d)], most methods perform well, given the ample temporal information available in the observations. Only minor distinctions are observed in reconstructing complex ground details. As contamination intensifies or involves rapid surface changes [S101–S102, e.g., snow coverage, Fig. 11(g)–(i)], the inherent limitations of physical methods become apparent. Both mean and min filters struggle with the rigid temporal constancy assumption, failing to capture actual land cover changes and resulting in discontinuities. Conversely, DL-based models utilize nonlinear learning to accommodate these variations, with GAN-based models and CDPCR generating clearer details.

In the most challenging scenarios characterized by dense and frequent contamination [S011–S111, Fig. 11(j)–(v)], differences in model performance become more apparent. Traditional filters show significant instability, fluctuating based on spectral diversity and often failing on surfaces lacking high-frequency details. Similarly, performance for most deep baselines (e.g., UnCRtainTS and U-TILISE) deteriorates markedly when clean references are absent, leading to severe artifacts and structural omissions. In contrast, CDPCR demonstrates superior robustness. By effectively leveraging the proposed cloud-deterministic diffusion and sampling strategies, it successfully retrieves coherent structures even when all observed frames are contaminated. We also observe that images reconstructed directly by the restoration network (CDPCR-R) closely match those obtained

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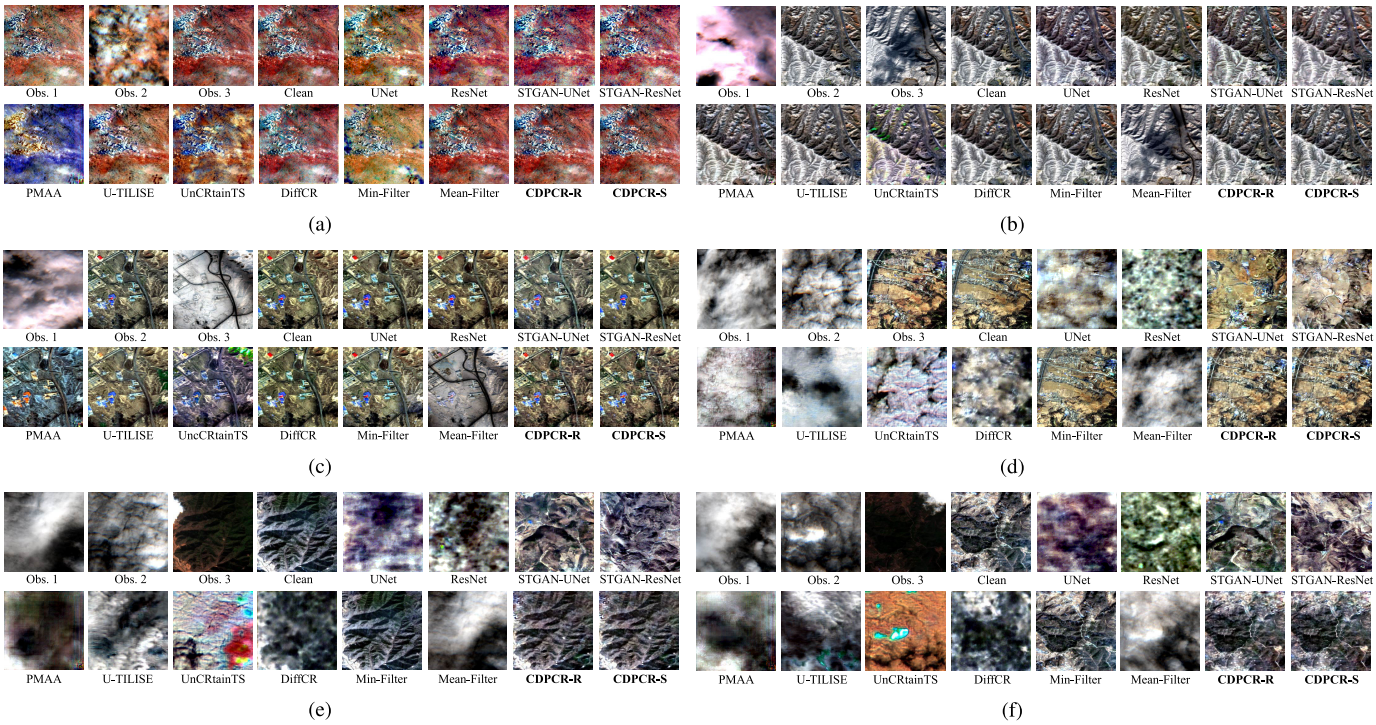


Fig. 5. Visualizations of model performance across different cloud contamination categories (S010-S111) on the SEN12MS-CR-TS dataset. (a) S010-type. (b) S101-type. (c) S102-type. (d) S110-type. (e) and (f) S111-type.

after sampling (CDPCR-S), which confirms the high fidelity of the trained restoration network and the validity of the proposed forward diffusion process. With increasing contamination severity, the improved sampler proves essential, as the residual-based updates (see Algorithm 2) yield more distinct improvements. Ultimately, while other methods struggle with the cloud-detail tradeoff, CDPCR consistently delivers realistic reconstructions, proving superior across diverse and complex atmospheric conditions.

For a more detailed examination of reconstruction quality across different models, we present enlarged visualizations of several representative scenes from different cloud contamination categories (S010-S111) in Figs. 5 and 6. Fundamentally, reconstruction quality is strongly influenced by cloud distribution and coverage. Traditional filter methods show clear limitations: the min filter performs well under favorable conditions, but rapidly degrades when faced with a lack of high-frequency detail and high spectral complexity [see Figs. 5(a) and 6(a)]. The mean filter, by contrast, exhibits limited effectiveness even in milder contamination [see Fig. 5(a)]. Deep generative models effectively remove clouds and cloud shadows, but they generally suffer from color deviations and distorted ground detail. We observe distinct flaws across baselines: UnCRtainTS and PMAA frequently introduce artifacts or fail to preserve the color and ground details [see Fig. 5(a), (b), and (f)]. Furthermore, in specific subregions, the traditional min filter and several DL baselines (e.g., UNet, STGAN-UNet, U-TILISE, and DiffCR) fail in spectral-temporal disambiguation, misinterpreting highly reflective features as clouds and creating unexpected artifacts [see bottom right of Fig. 5(a)]. ResNet and STGAN-ResNet exhibit instability, failing to accurately represent the entire scene. In contrast, CDPCR robustly distinguishes ground features from actual cloud contamination, ensuring high reconstruction fidelity to the underlying scenes.

As cloud contamination intensifies, most DL-based models suffer from more severe artifacts, blurring, and even catastrophic failure due to large, thick cloud occlusions [see Fig. 5(d)–(f)]. However, CDPCR yields more desirable results even with limited clean temporal information. The min filter also achieves moderate results, but its overall performance is unstable due to its simple temporal operation. Fig. 6 further illustrates magnified views of specific ground features (marked by yellow squares) from scenes with severe cloud contamination, allowing for a closer inspection of reconstruction details. Fig. 6(a) presents a challenging high-contamination, low-frequency scenario: here, CDPCR produces the most complete and accurate reconstruction of the ground features, providing superior recovery of subregion details. Fig. 6(b) further confirms this performance gap, exhibiting the most significant contrast under extreme weather conditions, where the observation suffers from varying levels of cloud contamination. The majority of baselines (STGANs, PMAA, and UnCRtainTS) fails to remove the clouds or struggle severely with reconstruction. CDPCR successfully removes the clouds, but its reconstruction may appear smoother than that of the min filter. However, this tradeoff is justified, as the min filter introduces unwanted artifacts in the left subregion occluded by persistent thick clouds.

In addition, in Fig. 6(a) and (b), we observed that results from DiffCR show noticeable checkerboard effects in some scenarios, which structurally impact the reconstructions. In contrast, CDPCR generates sharper details, both in the subregion and the entire image, demonstrating superior struc-

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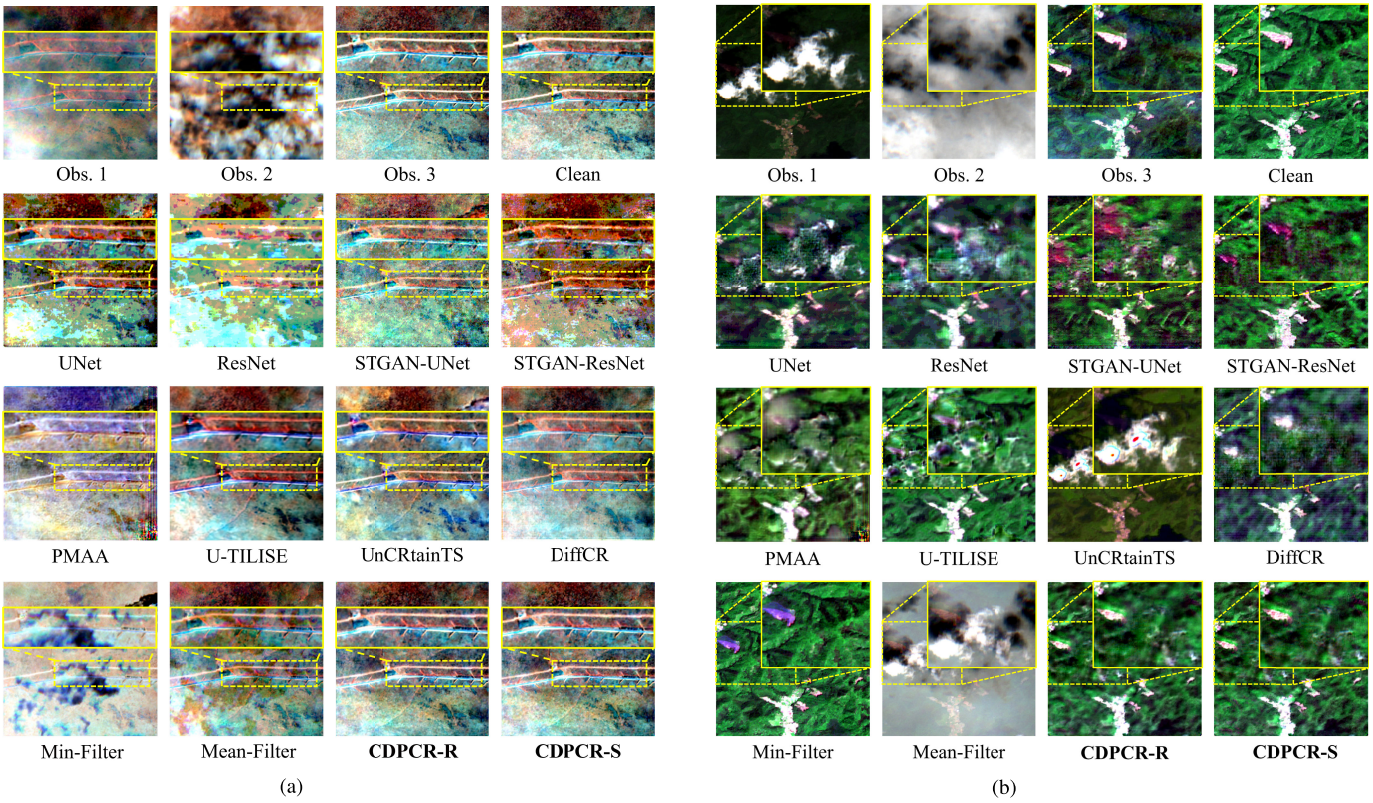


Fig. 6. Enlarged visualizations of reconstruction details from different models. Specific subregions (yellow squares) are zoomed-in for better comparison. (a) Scene 1. (b) Scene 2.

tural integrity. This performance gap is exacerbated as cloud contamination intensifies [see Figs. 5(e) and (f) and 6(b)]. This weakness is likely due to the inherent limitation of the standard Gaussian diffusion operator and the suboptimal sampling schedule. These factors make the existing diffusion models unsuitable for handling frequent, severe cloud contamination with limited temporal information. Overall, CDPCR maintains superior stability in scenes with complex cloud contamination and heterogeneous ground features, demonstrating the effectiveness and robustness of the proposed method in the real-world cloud removal task.

Based on Figs. 5, 6, and 11, we qualitatively analyze the performance of different models in scenarios with varying levels of cloud contamination. Among the physical methods, the min filter generally outperforms the mean filter, yielding reasonable results even in scenarios with intense cloud contamination. However, it deteriorates significantly in scenes with high spectral diversity—a specific condition where the mean filter offers slightly better stability, despite its overall blurriness. Fundamentally, these traditional filters face severe inherent limitations. They are constrained by conventional statistical operations, making them vulnerable to misleading features, particularly when high-frequency information is scarce. Moreover, their reliance on temporal constancy makes them unsuitable for observations with rapid land cover changes, leading to scientific inaccuracies and noticeable discontinuities in the final mosaicked results. In contrast to these rigid methods, CDPCR capitalizes on a specific training strategy and adaptive learning capabilities. By leveraging the cloud-deterministic diffusion and iterative sampling strategy, it consistently delivers robust reconstruction across diverse atmospheric conditions. Even in scenarios with frequent and severe occlusion, where minor degradation is unavoidable, CDPCR excels in preserving color accuracy, structural integrity, and detail retrieval, demonstrating its superiority over both physical filters and standard DL baselines.

Furthermore, such robust reconstruction capabilities are essential to meet the high temporal resolution requirements in many downstream applications [6], [75], which necessitate the complete reconstruction of observed sequences, ensuring continuity even amidst intermittent cloud cover. While conventional temporal compositing methods attempt to address this by simply replacing cloudy pixels with clear ones, they overlook phenological variations and are compromised by inconsistent atmospheric conditions. In reality, fully cloud-free imagery is scarce; instead, clear observations are frequently fragmented spatiotemporally. Consequently, relying on single-date references assumes ground feature stationarity, which is theoretically flawed given the dynamic nature of land cover. Moreover, identifying ideal clear references is often subjective and computationally inefficient. Therefore, effectively aggregating complementary information from adjacent temporal frames is essential for reconstruction. As cloud contamination intensifies, nonlinear generative models become crucial for synthesizing realistic details from limited available information. In this context, CDPCR achieves realistic S2S

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TABLE I
QUANTITATIVE EVALUATION OF CLOUD REMOVAL PERFORMANCE ON THE SEN12MS-CR-TS AND SEN2-MTC DATASETS

Model	Params (M)	Inf. Time (s)	SEN12MS-CR-TS (S010–S111)					Sen2-MTC (S111)				
			MSE ↓	MAE ↓	PSNR ↑	SSIM ↑	LPIPS ↓	MSE ↓	MAE ↓	PSNR ↑	SSIM ↑	LPIPS ↓
Mean Filter	0.0	0.005	0.0442	0.1615	24.9410	0.6723	0.1558	0.0047	0.0382	26.3325	0.8062	0.1690
Min Filter	0.0	0.004	0.0009	0.0112	33.0934	0.9227	**0.0477**	0.0007	0.0174	31.9047	0.7790	0.1097
UNet	31.0	0.026	0.0004	0.0129	38.5836	0.9271	0.0896	0.0005	0.0170	35.5198	0.8957	0.0641
ResNet	30.2	0.022	0.0008	0.0184	36.7335	0.9108	0.1029	0.0005	0.0183	34.6123	0.8805	0.0637
STGAN-UNet [42]	54.4	0.029	0.0006	0.0171	36.1549	0.9080	0.0642	0.0006	0.0187	34.5508	0.8770	0.0590
STGAN-ResNet [42]	38.2	0.024	0.0008	0.0182	36.4686	0.9001	0.0723	0.0006	0.0202	33.7904	0.8630	**0.0558**
UnCRtainTS [17]	12.5	0.058	0.0072	0.0577	21.5692	0.7215	0.1219	-	-	-	-	-
PMAA [41]	1.2	0.019	0.0007	0.0197	31.8300	0.8988	0.0996	0.0014	0.0229	28.8514	0.8376	0.1574
U-TILISE [70]	6.4	0.056	0.0004	0.0140	35.9057	0.9117	0.3267	0.0056	0.0580	26.0523	0.6001	0.2167
DiffCR [23]	25.7	0.310	0.0041	0.0472	23.9589	0.8960	0.0556	0.0080	0.0534	21.2699	0.8921	0.0867
EMRDM [76]	60.9	1.051	0.0008	0.0182	31.1604	0.9057	0.0912	0.0011	0.0219	30.0518	0.9017	0.1076
CDPCR	56.6	0.681	**0.0003**	**0.0108**	**39.6716**	**0.9352**	0.0599	**0.0003**	**0.0123**	**37.3066**	**0.9288**	0.0770

recovery, successfully reconstructing complete observations with complex occlusion scenarios on the SEN12MS-CR-TS dataset (see Fig. S3 in the Supplementary Material).

2) Quantitative Evaluation Across Varying Cloud Contamination Levels: To further evaluate model performance, we conduct quantitative assessments comparing CDPCR with several SOTA baselines. To ensure fair comparison, all baseline models were retrained on the dataset synthesized by our proposed simulation protocol, following their official implementation details. For quantitative evaluation, the contaminated input $\mathbf{X}_S$ is synthesized from clean backgrounds $\mathbf{X}_0$ using the cloud-deterministic diffuser $\mathcal{D}$. To mitigate discrepancies caused by land cover changes, temporal alignment is maintained by pairing the GT $\mathbf{X}_0$ and the observed sequence based on their closest temporal proximity. This setup allows the model to aim for the recovery of the original underlying images ($\mathbf{X}_0$), which serve as the explicitly available GT for reliable metric calculation. Crucially, all models are evaluated on the same synthetic test set to ensure a consistent and realistic contamination standard.

Table I presents the quantitative evaluation of cloud removal performance on the SEN12MS-CR-TS dataset [50]. Among physical methods, the min filter attains the best LPIPS performance, reflecting better alignment with human perception and suggesting the potential of temporal information extraction. Furthermore, it demonstrates superior stability compared with the mean filter, positioning it as a promising candidate for coarse cloud processing. Nevertheless, these physical models lag behind DL-based methods in terms of mse, PSNR, and SSIM metrics. Specifically, the U-TILISE model demonstrates robust performance, achieving the second-lowest mse among all compared methods, surpassed only by CDPCR. Similarly, the UNet model delivers reliable performance, underscoring the efficiency of its concise architecture for cloud removal. Among diffusion-based baselines, DiffCR exhibits strong structural integrity—reflected by high SSIM and competitive LPIPS scores—indicating its capability to preserve major outlines and global structures. However, this advantage is compromised by a lower PSNR, as the metric is highly sensitive to the noticeable checkerboard artifacts, which manifest as regular, high-frequency pixel-level errors. In contrast, EMRDM delivers noticeably improved performance over DiffCR in terms of PSNR and SSIM, reflecting the effectiveness of its advanced generative mechanisms. Nevertheless, the proposed CDPCR consistently outperforms the competing methods on most evaluation metrics, achieving the highest PSNR of 39.6716 dB and SSIM of 0.9352. These results demonstrate its superior capability in removing clouds while preserving pixel-level accuracy and structural fidelity. Although its LPIPS score is comparable to that of DiffCR, CDPCR demonstrates a stronger overall balance between pixel-level accuracy and structural fidelity across the evaluated metrics.

Given unpredictable atmospheric conditions, cloud-free information is not always available. The difficulty of reconstruction increases as the amount of clean information in the observed sequences diminishes. To investigate the stability and performance fluctuation across different conditions, we utilized the SEN12MS-CR-TS dataset [50], which provides abundant observations featuring diverse cloud distributions, coverages, and contamination severities, allowing for a comprehensive evaluation across varying degradation levels. Detailed quantitative assessments across the full spectrum of mild to severe cloudy scenes are presented in Supplementary Tables S1–S7. These results demonstrate CDPCR's robust performance across these scenarios. It achieves a PSNR exceeding 49 dB in favorable S010 and S110-a scenes, which confirms its high reconstruction fidelity. Notably, in the most difficult scenarios (S111-type), CDPCR maintains stability, achieving the best metric values across mse, PSNR, and SSIM, though it trails slightly in MAE and LPIPS metrics (Table S7). Analysis of the baselines also reveals notable variations: DiffCR obtains favorable LPIPS scores when cloud contamination is milder, whereas U-TILISE demonstrates more stable performance across metrics and scenes. Furthermore, the min filter and UNet exhibit reasonable performance in different scenarios, confirming the efficacy of these simple algorithms under appropriate conditions for cloud removal.

Fig. 7 visually summarizes model performance variations using the PSNR metric. While models generally display a downward trend as contamination intensifies, the results

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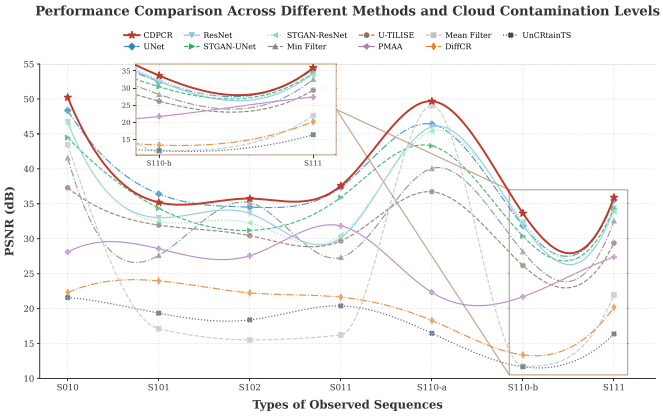


Fig. 7. Performance variations of cloud removal models across varying cloud contamination levels.

exhibit fluctuations, suggesting performance is simultaneously influenced by cloud layers and underlying ground complexity, as demonstrated by the metric variations observed between S110-a and S110-b types. Furthermore, the temporal position of cloud-free information significantly impacts results: reconstructions of S011 outperform those of S101 and S110-b (S011 > S101 > S110-b). Under mild contamination, baseline curves closely follow CDPCR (represented by the red curve). However, as contamination intensifies, CDPCR maintains a gentler decline and greater stability. To explicitly illustrate this superiority in severe scenarios, an inset magnified view is provided for the heavily contaminated S110-b and S111 sequences. As highlighted in this zoomed-in region, CDPCR maintains a distinct, consistent performance advantage over competing methods. Crucially, it achieves the most robust results even in the most challenging S111-type scenes.

Our analysis confirms that insufficient cloud-free information degrades reconstruction quality, underscoring the important role of auxiliary temporal data in maintaining image fidelity. Nevertheless, the comprehensive performance demonstrates CDPCR's effective and robust capability for cloud removal and image reconstruction across scenarios with varying levels of cloud contamination. Beyond atmospheric conditions, we also observed that model performance is significantly influenced by the underlying ground features. While utilizing diversified input data is beneficial for generalization, it may simultaneously introduce irrelevant or conflicting features that hinder reconstruction accuracy. Therefore, finding the optimal balance between data diversity and reconstruction quality remains a critical direction for future exploration.

C. Experimental Results on Sen2-MTC

1) Visualization of Reconstructions: To further evaluate the reconstruction performance in S111-type scenes, we conduct experiments on the Sen2-MTC dataset [14]. In contrast to the SEN12MS-CR-TS dataset [50], Sen2-MTC features images with a consistently severe contamination level within each observation, characterized by prevalent cloud and shadow effects. Visual comparisons are provided in Figs. S5 and S6. On this challenging dataset, the min filter suffers significant

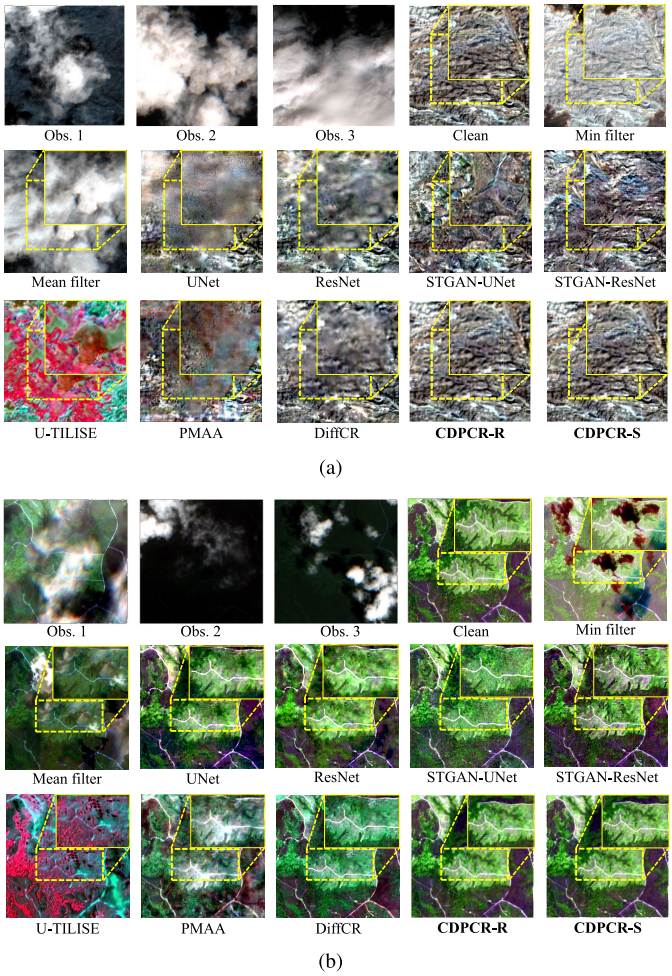


Fig. 8. Enlarged reconstruction details from different models on the Sen2-MTC dataset. Specific subregions are zoomed-in for better comparison. (a) Scene 1. (b) Scene 2.

performance degradation. This is primarily due to the prominent cloud shadows and limited temporal references, which undermine the method's effectiveness. The mean filter also fails to effectively remove clouds or reconstruct underlying details. Among DL-based methods, U-TILISE shows suboptimal performance, exhibiting serious color deviations and cloud residues. This is likely due to its dependency on all 13 bands, a requirement not met by the four-band configuration of Sen2-MTC. While STGAN models outperform UNet and ResNet in detail recovery, they remain visually inferior to DiffCR and CDPCR. Overall, CDPCR outperforms all compared models, providing more thorough cloud removal, more accurate color restoration, and superior detail reconstruction. Complete S2S reconstructions generated by CDPCR are illustrated in Fig. S6 of the Supplementary Material.

Fig. 8 presents enlarged visualizations of selected regions for detailed analysis. In Fig. 8(a), filtering methods fail to remove the clouds, and U-TILISE restores the image but exhibits severe color deviations. The remaining DL-based methods successfully reconstruct the scene, with STGAN, DiffCR, and CDPCR particularly excelling in generating a clear and complete reconstruction of the road feature. However,

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the majority of baselines (UNet, ResNet, STGAN, PMAA, and DiffCR) struggle significantly to recover the green-toned feature, showing only vague preservation and lacking accurate structure and details. Conversely, CDPCR thoroughly removes cloud contamination and reconstructs the image with complete and realistic details. In the scene of Fig. 8(b), the proposed CDPCR delivers superior reconstruction, demonstrating outstanding structural integrity and realistic detail recovery (magnified view). The effectiveness of the sampling process (CDPCR-S) is evident, as results exhibit CDPCR-R. Overall, the visualizations in Figs. 8, S5, and S6 qualitatively validate CDPCR's robust capability against the frequent and severe cloud and shadow contamination present in the Sen2-MTC dataset. The subsequent quantitative evaluation further assesses the compared models.

2) Quantitative Evaluation: We conduct a quantitative evaluation to assess cloud removal performance on the Sen2-MTC dataset [14]. Table I presents the quantitative metrics for the compared models, providing an objective assessment of reconstruction quality. As shown in Table I, the min filter suffers significant degradation on the Sen2-MTC dataset, whereas UNet and STGAN models demonstrate robust performance. Notably, STGAN exhibits consistent stability, delivering reliable results across both the SEN12MS-CR-TS and Sen2-MTC datasets. Among the deep generative baselines, DiffCR maintains strong SSIM and LPIPS scores, and PMAA outperforms U-TILISE, likely due to the latter's limitations under restricted spectral channels. EMRDM outperforms DiffCR, achieving a competitive SSIM score of 0.9017 and demonstrating strong performance on the challenging S111-type scenes. In comparison, CDPCR achieves the best performance across the primary objective metrics. Although its LPIPS score is slightly lower than that of STGAN, CDPCR still outperforms DiffCR on this dataset, demonstrating its strong adaptability and overall robustness for cloud removal.

In this study, we conducted extensive experiments across multiple datasets and scenarios, tracking the performance stability of various models under varying cloud coverages, distributions, morphologies, and heterogeneous land cover types. This allows for a rigorous assessment of the capability and robustness of the proposed CDPCR. The results demonstrate that the proposed framework outperforms compared SOTA baselines on both the SEN12MS-CR-TS and Sen2-MTC datasets, providing reliable S2S reconstructions of optical time-series affected by diverse, complex, and even temporally persistent cloud contamination (including thick clouds and shadows). This success can be attributed to the novel cloud-deterministic diffusion process designed in CDPCR, which gradually intensifies natural cloud contamination in the background. This approach broadens training exposure to varying cloud patterns and intensities, enabling robustness against complex cloud degradations. An improved sampling rule is then used for image refinements, which allows the model to progressively approach realistic data distributions by iteratively alternating between the forward and backward processes. Overall, CDPCR adaptively tackles the inherent complexities introduced by varied cloud contamination and diverse ground features.

D. Model Complexity and Computational Efficiency Analysis

Table I presents a unified quantitative comparison of parameter footprint (Params), inference latency (Inf. Time), and reconstruction performance across all baseline methods. While single-pass feed-forward baselines such as UNet (31.0 M), ResNet (30.2 M), and lightweight models like PMAA (1.2 M) possess smaller model scales, their representation capacity remains limited when addressing complex spatio-temporal cloud contamination. In contrast, within the generative diffusion paradigm, CDPCR maintains a highly parameter-efficient footprint (56.6 M). This scale is significantly more compact than SAR-optical and multitemporal diffusion frameworks like ST-Diffusion (110.0 M) [25] and comparable to generative architectures such as STGAN-UNet (54.4 M) and EMRDM (60.9 M). This modest increase in parameters over basic CNNs represents a strategic design choice that provides sufficient modeling capacity to capture intricate temporal dependencies and spectral feature interactions.

Due to their iterative sampling process, diffusion-based models generally incur higher inference latency than conventional feed-forward restoration models. In contrast, most feed-forward baselines achieve < 0.06 s inference times under the evaluated setting. CDPCR requires an average of 2.044 s for a single multitemporal sequence (dimension of $256 \times 256 \times 3 \times 3$) on an NVIDIA RTX 3090 GPU, corresponding to approximately 0.681 s/image. DiffCR exhibits a faster execution time (0.310 s), yet delivers lower reconstruction performance than CDPCR under identical experimental settings. Overall, CDPCR provides a favorable tradeoff between model complexity, inference efficiency, and reconstruction fidelity. It delivers competitive fidelity across both datasets (39.67-dB PSNR on SEN12MS-CR-TS and 37.31-dB PSNR on Sen2-MTC) with manageable parameter size and reasonable inference speed. These results indicate that CDPCR has strong potential for practical satellite image reconstruction applications.

V. Discussion

In this section, we perform ablation studies on CDPCR to evaluate the influence of its core components and design choices on model performance. Specifically, six targeted ablation experiments are conducted to systematically investigate key algorithmic and architectural factors. Furthermore, we discuss current limitations and outline potential directions for future research. Sections V-A–V-C detail these assessments and critical findings.

A. Ablation Study

1) Effect of Different Cloud Degradation Schedules: To verify the necessity and superiority of our proposed cloud-deterministic diffusion paradigm, we conduct a rigorous controlled experiment by comparing it against the standard Gaussian diffusion baseline and several degradation variants.

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TABLE II
QUANTITATIVE ASSESSMENT OF CLOUD REMOVAL PERFORMANCE UNDER DIFFERENT DIFFUSION PROCESSES AND SCHEDULES

Model	Diffusion Schedule	Mask Type (M)	MSE	MAE	PSNR	SSIM	LPIPS
Standard	$\mathbf{X}_s = \sqrt{1-w_s}\mathbf{X}_0 + \sqrt{w_s}\epsilon$	No mask	0.0014 *0.0014*	0.0296 *0.0296*	29.9730 *29.9730*	0.8661 *0.8661*	0.2619 *0.2619*
CDPCR	$\mathbf{X}_s = (1-\sqrt{w_s})\mathbf{X}_0 + \sqrt{w_s}\mathbf{Y}$	**No mask**	**0.0003** *0.0003*	**0.0108** *0.0108*	**39.6716** *39.6716*	0.9352 *0.9352*	**0.0599** *0.0599*
CDPCR_1	$\mathbf{X}_s = \sqrt{1-\sqrt{w_s}}\mathbf{X}_0 + \sqrt[4]{w_s}\mathbf{Y}$	No mask	0.0003 *0.0003*	0.0112 *0.0112*	39.2726 *39.2726*	**0.9372** *0.9372*	0.0677 *0.0677*
CDPCR_2	$\mathbf{X}_s = (1-\sqrt{w_s})\mathbf{X}_0 + \sqrt{w_s}\mathbf{Y} \odot \mathbf{M}$	IHOT	0.0051 *0.0003*	0.0212 *0.0103*	33.6552 *39.6649*	0.8970 *0.9354*	0.1467 *0.0634*
CDPCR_3	$\mathbf{X}_s = (1-\sqrt{w_s})\mathbf{X}_0 + \sqrt{w_s}\mathbf{Y} \odot \mathbf{M}$	S2PixelCDetector	0.0168 *0.0001*	0.0451 *0.0052*	29.2062 *43.4396*	0.8250 *0.9698*	0.2141 *0.0357*
CDPCR_4	$\mathbf{X}_s = (1-\sqrt{w_s})\mathbf{X}_0 + \sqrt{w_s}\mathbf{Y} \odot \mathbf{M}$	Low-frequency	0.0232 ***1.1353E-05***	0.0843 ***0.0011***	25.0387 ***68.2840***	0.7607 ***0.9978***	0.2327 ***0.0031***

Note: "Standard" represents the baseline configuration employing the standard Gaussian diffusion process. For a controlled and fair comparison, it incorporates the identical restoration network architecture as the proposed CDPCR. Values in roman and italic fonts represent results on real-world and simulated datasets, respectively.

To ensure a strictly fair evaluation, all compared schedules utilize the identical restoration network architecture.

First, comparing the standard Gaussian baseline and CDPCR underscores the decisive role of the forward degradation schedule in cloud removal. The standard Gaussian diffusion baseline, which degrades the clean image by injecting pure random noise, exhibits limited performance relative to the other configurations. Specifically, it obtains a PSNR of 29.9730 dB and an SSIM of 0.8661, accompanied by a relatively high perceptual distortion (LPIPS of 0.2619). This severe performance drop stems from the fundamental mismatch between Gaussian noise and natural cloud contamination, as motivated in the introduction. While Gaussian noise introduces pixel-independent, high-frequency perturbations, natural clouds possess complex macro-spatial structures, continuous boundaries, and varying thicknesses. Apart from the mismatch in degradation patterns, the unconditional generation mechanism under this paradigm inherently struggles to achieve ideal cloud removal and restoration results. Consequently, pretraining the network to remove nonphysical Gaussian noise fails to equip it with the capability to capture and eliminate macro-structural cloud covers in real-world scenes, leading to compromised restoration.

In contrast, the proposed CDPCR incorporates the cloud-deterministic diffusion schedule (4), directly utilizing the natural cloud-contaminated image $\mathbf{Y}$ as the forward degradation source. By doing so, CDPCR achieves a remarkable performance leap under the exact same network capacity, boosting the PSNR to 39.6716 dB and reducing the LPIPS to 0.0599. This substantial performance gap demonstrates that aligning the forward diffusion process with the true physical and morphological characteristics of cloud contamination significantly eases the learning difficulty of the restoration network. This controlled comparison strongly validates that standard Gaussian diffusion is intrinsically less suitable for cloud removal, and matching the forward degradation with actual cloud features is vital.

Furthermore, we explore other variants within the cloud-deterministic framework to evaluate the impact of different blending trajectories and mask priors. CDPCR_1 adopts an alternative nonlinear schedule, which yields a slight performance degradation compared with the standard CDPCR. This indicates that a well-paced linear decay of clean image features combined with cloud injection provides a more stable optimization path. On the other hand, CDPCR_2, CDPCR_3, and CDPCR_4 integrate different cloud mask constraints (IHOT, S2PixelCloudDetector, and low-frequency components) into the diffusion process. To assess the impact of mask guidance, we present results from both simulated and real-world experiments in Table II. Interestingly, while these mask-guided variants show exceptional performance on the simulated experiments—particularly CDPCR_4, which achieves a remarkable PSNR of 68.2840 dB by leveraging low-frequency priors—their performance drops significantly when applied to real-world datasets. This discrepancy arises because masks can perfectly capture simulated cloud regions, whereas in real-world scenarios, the complex, ambiguous boundaries of natural cloud contamination make accurate mask extraction extremely challenging. The resulting mask estimation errors heavily propagate into the restoration stage, leading to suboptimal generalization. Conversely, the standard CDPCR operates successfully without explicit mask dependencies, demonstrating the most robust and stable performance in practical, real-world cloud removal deployment.

2) Effect of the Number of Forward Diffusion Steps: The diffusion steps specify the number of degraded states between the original background $\mathbf{X}_0$ and the final distribution $\mathbf{X}_S$. The total number of steps directly governs the size of the degradation interval, where a sufficient count is essential to ensure a smooth and gradual cloud degradation process. Thus, we conduct a quantitative analysis of the impact of the total diffusion steps on model performance (see Table III).

Experimental observations show that model performance does not increase indefinitely with the number of diffusion

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TABLE III
QUANTITATIVE ASSESSMENT OF CDPCR VARIANTS WITH DIFFERENT FORWARD DIFFUSION STEPS

Model	Diff. Steps	MSE	MAE	PSNR	SSIM	LPIPS
CDPCR	85	0.0003	0.0116	39.0225	0.9334	0.0615
	90	**0.0003**	**0.0106**	**39.6787**	**0.9374**	**0.0588**
	95	0.0003	0.0116	39.0875	0.9364	0.0589
	100	0.0003	0.0108	39.6716	0.9352	0.0599
	105	0.0003	0.0110	39.5077	0.9308	0.0635
	110	0.0003	0.0112	39.2657	0.9331	0.0624
	120	0.0004	0.0119	39.0261	0.9323	0.0634
	130	0.0003	0.0109	39.4733	0.9345	0.0602
	140	0.0003	0.0113	39.2982	0.9336	0.0623
	150	0.0003	0.0111	39.3183	0.9373	0.0603
	160	0.0004	0.0117	39.3217	0.9329	0.0620
	170	0.0003	0.0113	39.3555	0.9323	0.0630
	180	0.0003	0.0111	39.4704	0.9364	0.0590
	190	0.0003	0.0114	39.1278	0.9319	0.0632
	200	0.0003	0.0112	39.1364	0.9322	0.0639

steps. Specifically, Table III reveals a performance peak at 90 steps, with the 100-step setting remaining highly competitive and comparable. However, extensive robustness testing across diverse unseen scenarios revealed that the 90-step configuration occasionally introduces potential qualitative artifacts. This phenomenon suggests that a restricted number of diffusion steps might limit the diversity of degradation forms synthesized during the training phase. Consequently, the model may struggle to learn robust and stable reconstruction patterns to generalize across complex and volatile real-world atmospheric conditions. These findings further underscore that maintaining an appropriate level of complexity and diversity in the degradation process is paramount for ensuring the stability of DL-based restoration models. Therefore, considering both reconstruction precision and cross-scene applicability, we selected 100 steps as the standard configuration for CDPCR. This choice is not an isolated numerical peak; rather, it marks the entry point into a stable performance plateau. As evidenced in Table III, performance remains consistently robust as steps increase from 100 to 200. By prioritizing cross-scene robustness over marginal gains in numerical scores, this configuration effectively mitigates generalization vulnerabilities under diverse or nonstationary cloud contamination, ensuring the reliability of CDPCR in practical remote sensing applications.

3) Effect of Different Restoration Network Architectures: To investigate the impact of model capacity and architectural design on the performance of the proposed CDPCR paradigm, we evaluated its effectiveness and generalizability across different restoration backbones. Specifically, we evaluated the performance of CDPCR with three distinct restoration backbones: a baseline U-Net, a lightweight U-Net, and an unconditional DiffCR-style architecture. Notably, since the reverse process of CDPCR operates directly on the input sequences, we did not directly adopt DiffCR's explicit conditional concatenation pathway. Instead, we incorporated its advanced spatial block design to construct an unconditional DiffCR-style architecture. The quantitative results, including model complexity (Parameters and FLOPs) and restoration performance (PSNR and SSIM), are summarized in Table IV. To ensure a fair comparison, all models were evaluated under both the standard Gaussian diffusion paradigm and the proposed cloud-deterministic diffusion paradigm on the SEN12MS-CR-TS dataset.

TABLE IV
QUANTITATIVE COMPARISON OF DIFFERENT BACKBONE ARCHITECTURES AND DIFFUSION PARADIGMS

Architecture	Params (M)	FLOPs (G)	Metrics	
			PSNR ↑	SSIM ↑
Original U-Net	56.62	14410.25	39.6716 *29.9730*	0.9352 *0.8661*
Lightweight U-Net	14.47	3784.01	**41.3271** *24.8962*	**0.9591** *0.7432*
Uncond. DiffCR-style	19.60	2444.57	40.8387 *23.8691*	0.9507 *0.6658*

Note: For each architecture, the metric values in normal font (top row) represent the results of the proposed cloud-deterministic diffusion paradigm, while the values in *italics* (bottom row) correspond to the standard Gaussian diffusion paradigm.

The proposed cloud-deterministic paradigm consistently and significantly outperforms the standard Gaussian paradigm across all tested architectures. In contrast, the standard Gaussian diffusion paradigm exhibits compromised performance, primarily due to the inherent mismatch of the degradation noise and the lack of conditional image guidance. Under the cloud-deterministic paradigm, the PSNR improves by approximately 10–17 dB across these backbones compared with their Gaussian counterparts, suggesting that the effectiveness of the proposed method is not tied to a specific network. Notably, even when implemented on a Lightweight U-Net (with a 75% reduction in parameters and a 73% reduction in FLOPs compared with the original U-Net), the CDPCR paradigm achieves a remarkable PSNR of 41.32 dB. This suggests that the proposed diffusion paradigm simplifies the learning objective, allowing even compact models to achieve superior restoration quality. Furthermore, the cloud-deterministic paradigm also shows exceptional performance on the DiffCR-style architecture (40.83 dB), further confirming its broad applicability. These results demonstrate that the proposed CDPCR is a robust, architecture-agnostic diffusion paradigm that provides substantial performance gains across a wide range of model scales and designs.

4) Effect of Different Sampling Algorithms: The sampling process is a crucial procedure in diffusion models, which provides the rule for restoration updates by iteratively alternating between the restoration network ($\mathcal{R}$) and the diffusion operator ($\mathcal{D}$). We quantitatively assess the impact of different sampling algorithms on model performance by applying them to reverse natural cloud contamination. The improved sampler (A2), as detailed in Section III-B2, is designed to capture more high-frequency information, which is critical for handling severe cloud contamination. To assess the efficiency of A2 compared with A1 (standard sampler) and validate its role in the CDPCR process, we devised a Hybrid strategy. This strategy dynamically adjusts the proportion of A2 utilized based on a set high frequency threshold (τ, representing the high-frequency ratio

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TABLE V
ASSESSMENT OF TRANSFERABILITY AND ROBUSTNESS ACROSS DIFFERENT DATA DISTRIBUTIONS, SENSORS, AND TRAINING STRATEGIES

Model	Setting	Train Dataset	Test Dataset	Strategy	MAE	PSNR	SSIM
CDPCR	Intra-sensor	SEN12MS-CR-TS	SEN12MS-CR-TS	Direct	0.0108	39.6716	0.9352
		SEN12MS-CR-TS	Sen2-MTC	Direct	0.0127	36.0037	0.9182
		Sen2-MTC	Sen2-MTC	Direct	0.0123	37.3066	0.9288
	Cross-sensor	SEN12MS-CR-TS	AllClear-L8/9	Direct	0.0184	35.7431	0.9371
		SEN12MS-CR-TS + AllClear-L8/9	**AllClear-L8/9**	**Fine-Tune**	**0.0147**	**37.8137**	**0.9685**

Note: SEN12MS-CR-TS [50] and Sen2-MTC [14] are derived from Sentinel-2 imagery; AllClear-L8/9 [77] is derived from Landsat 8/9 imagery. "Direct": Direct training without fine-tuning. Trivial samples (SSIM $>$ 0.99) were excluded to ensure a fair evaluation of cloud removal capability.

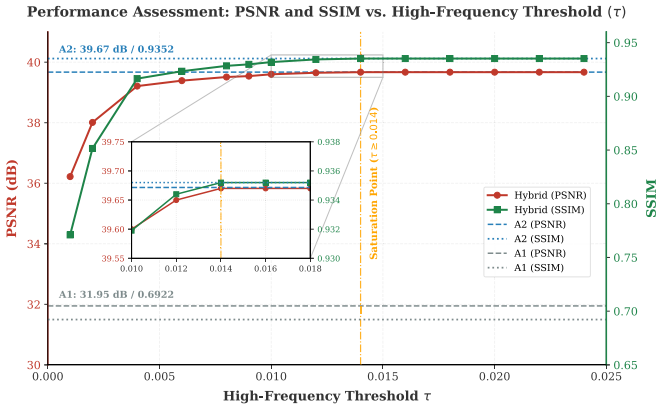


Fig. 9. Performance assessment of CDPCR using different sampling methods.

in $\mathbf{X}_{s-1}$). Specifically, the sampler switches back A1 when the proportion of high-frequency components in $\mathbf{X}_{s-1}$ reaches the threshold τ; otherwise, it utilizes A2 as the updating rule for sampling.

By varying the τ value, we controlled the proportion of A2 usage, observing that reconstruction performance improves rapidly before plateauing (see Fig. 9). As highlighted by the saturation line ($\tau \geq 0.014$) and the zoomed-in inset view, both PSNR and SSIM metrics smoothly converge toward the upper performance bound established by the pure A2 sampler (39.67 dB and 0.9352), far outperforming the baseline using *A*1 alone (31.95 dB and 0.6922). This superior SSIM also reflects the ability of the A2 sampler to suppress common generative hallucinations by prioritizing structural consistency via its residue-care update rule. Given the added complexity of the Hybrid strategy and the fact that it yields no further gain beyond the saturation threshold, we ultimately selected the pure improved algorithm (A2) as the primary sampling method for CDPCR. Complete metric evaluations are included in Table S10 of the Supplementary Material.

5) Analysis of Cross-Domain Generalization: To comprehensively evaluate the generalization capability of CDPCR, we conducted experiments under two distinct scenarios: 1) *Intrasensor distribution shifts*, where the model trained on SEN12MS-CR-TS [50] is tested on Sen2-MTC [14] with different cloud contamination patterns; and 2) *Cross-sensor domain shifts*, where the model is transferred to Landsat 8/9 imagery from the AllClear dataset [77], referred to as AllClear-L8/9. The quantitative results are summarized in Table V.

a) Robustness to distribution shifts (intrasensor): Although both SEN12MS-CR-TS and Sen2-MTC originate from Sentinel-2, they exhibit distinct data distributions: the former captures the variability of natural cloud contamination, while the latter focuses on more frequent and intense degradation (S111-type). As shown in the first section of Table V, when directly applied to the unseen Sen2-MTC dataset (Row 2), the model achieves a high PSNR of 36.00 dB and an SSIM of 0.9182. This performance remains highly competitive with the domain-specific upper bound (Row 3), exhibiting only a marginal decline (e.g., a gap of approximately 1.3 dB in PSNR and 0.01 in SSIM). Considering the significant distribution shift—where Sen2-MTC features consistently severe S111-type contamination compared with the variable conditions in the source domain—this narrow performance gap confirms that CDPCR learns generalized, dataset-agnostic representations rather than overfitting to specific cloud patterns. This underscores the model's robustness and generalization capability to handle severe degradation scenarios.

b) Transferability across sensors (cross-sensor): To explore the generalization to sensors with different radiometric properties, we evaluated the model on the AllClear-L8/9 dataset. When directly transferring the model trained on SEN12MS-CR-TS to the AllClearr-L8/9 dataset (without fine-tuning), we observed a notable contrast: although the PSNR metric decreased compared with intrasensor testing, the model achieved a remarkably high SSIM of 0.9371, surpassing even the intrasensor results (Row 4). This phenomenon can be attributed to the lower spatial resolution of Landsat (30 m) compared with Sentinel-2 (10 m), which naturally yields higher SSIM scores due to reduced high-frequency variance. Nevertheless, the high structural fidelity—despite the radiometric domain gap and the drop in PSNR—demonstrates that CDPCR possesses inherent cross-sensor generalization capabilities even before adaptation. Furthermore, applying a fine-tuning strategy significantly boosts performance, improving PSNR from 35.74 to 37.81 dB and SSIM from 0.9371 to 0.9685. These results confirm that CDPCR effectively learns sensor-independent structural priors that can be efficiently adapted to other optical sensors via transfer learning, validating its potential for multisource remote sensing applications.

6) Analysis of Temporal and Spectral Settings: The proposed CDPCR model is flexible, allowing easy adjustment of

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TABLE VI
QUANTITATIVE ASSESSMENT OF CDPCR PERFORMANCE UNDER VARYING INPUT CONFIGURATIONS

Model	Bands	Frames	MAE	PSNR	SSIM	LPIPS
Part I: Influence of Temporal Sequence Length (B = 3)						
CDPCR	3	1	0.0117	39.7073	0.9166	0.0692
	3	2	0.0117	39.7528	0.9120	0.0768
	3	3	0.0108	39.6716	0.9352	0.0599
	3	4	0.0088	41.0264	0.9538	0.0424
	3	5	**0.0079**	**42.0274**	**0.9565**	**0.0378**
Part II: Influence of Spectral Band Configurations (T = 3)						
CDPCR	3	3	0.0108	39.6716	0.9352	0.0599
	7	3	**0.0096**	**40.3738**	**0.9525**	**0.0401**
	13	3	0.0104	39.3696	0.9390	0.0682

input sequence length to meet practical demands. Table VI summarizes the results of this ablation study, where we utilize RGB images for training and vary the temporal length (T) from 1 to 5 images. The final four rows present the performance of configurations ranging from 3 to 13 bands (all with three time frames), serving as spectral fidelity benchmarks. To ensure optimal feature extraction and reconstruction within the respective spectral-temporal space, the model is specifically trained for each input configuration.

As demonstrated in Table VI, CDPCR's performance gradually improves as the temporal length increases. Crucially, when the number of input images reaches four, the observations provide more temporal clues about the underlying features, enabling the model to achieve a PSNR value exceeding 40 dB. Notably, CDPCR performs effectively even with minimal temporal data. The model even achieves higher PSNR values with 1–2 input frames than with the three-frame setting, although other structural metrics are substantially improved as T increases to 3. In addition, expanding the spectral input from 3 to 7 bands yields a substantial performance gain, confirming CDPCR's robust capability in high-dimensional feature learning. While the 13-band configuration maintains superior physical consistency (MAE/SSIM), it exhibits a subtle tradeoff in PSNR and LPIPS. This variation is primarily attributed to the increased heterogeneity across the full spectrum of Sentinel-2 bands and the heightened complexity of ordinary differential equation (ODE) trajectory evolution within expanded state spaces. Ultimately, CDPCR reaches an optimal balance of information gain at a specific spectral density, effectively prioritizing global structural integrity over extreme pixel-level fitting in complex spectral scenarios. These results underscore the flexibility of CDPCR, where the core architecture and diffusion logic remain unified across diverse data conditions.

B. Adaptability and Performance on High-Resolution Images

To further assess the adaptability and generalization of CDPCR in spatially challenging scenarios and mono-temporal settings, we evaluate its performance on high-resolution cloud removal using the CUHK-CR dataset [78]. The CUHK-CR dataset is a single-temporal dataset consisting of ultrahigh-resolution optical imagery captured by China's Jilin-1 (JL-1) commercial satellite series, boasting a spatial resolution of 0.5 m. Although the original satellite sensors acquire four spectral bands [RGB and near-infrared (NIR)], our experiments specifically utilize the three RGB channels. The benchmark dataset encompasses two distinct subsets: CUHK-CR1 (thin/hazy cloud contamination) and CUHK-CR2 (thick/opaque cloud contamination).

TABLE VII
QUANTITATIVE COMPARISON AND COMPUTATIONAL COMPLEXITY ON THE HIGH-RESOLUTION CUHK-CR DATASET (T = 1)

Model	Params (M)	Inf. Time (s)	PSNR ↑	SSIM ↑	LPIPS ↓
EMRDM [76]	60.92	1.401	30.2931	0.9046	0.1058
CDPCR (dim=32)	**14.47**	**1.091**	**34.5378**	**0.9310**	**0.0534**

For rigorous evaluation under varying cloud-contamination levels, we merged CUHK-CR1 and CUHK-CR2 and cropped their images from 512 × 512 × 3 to 256 × 256 × 3 subpatches. The consolidated dataset was randomly shuffled and repartitioned into training, validation, and test sets according to an 8:1:1 ratio. Since CUHK-CR is a monotemporal dataset, the input temporal dimension of CDPCR was configured to T = 1, yielding an input tensor shape of 256 × 256 × 3 × 1. Given the reduced input channels in single-temporal processing compared with multitemporal sequences, CDPCR was configured with a reduced feature channel dimension of dim = 32. For comparison, we evaluate against EMRDM [76], a representative diffusion-based cloud removal model applicable to both monotemporal and multitemporal settings, under identical input configurations and evaluation protocols.

As reported in Table VII, CDPCR exhibits solid quantitative performance and efficiency on high-resolution single-temporal imagery. Specifically, CDPCR achieves a PSNR of 34.5378 dB, an SSIM of 0.9310, and an LPIPS score of 0.0534, showing consistent improvements over EMRDM. These metrics indicate the model's ability to retain spatial fidelity and perceptual clarity in single-frame and high-resolution reconstruction. Regarding computational efficiency, under the dim = 32 setting tailored for reduced input channels, CDPCR maintains a compact parameter size of 14.47 M and an inference latency of 1.091 s/image.

The visual comparison in Fig. 10 further illustrates the reconstruction capability of CDPCR under high-resolution conditions and different levels of cloud contamination. As observed in the zoomed-in regions, while EMRDM effectively attenuates cloud cover, it struggles to recover high-frequency spatial structures under thick cloud regions, often generating blurry textures and severe color distortion. In contrast, CDPCR removes complex cloud occlusions, completely eliminating residual haze while reconstructing clearer and more comprehensive building boundaries and surface features that closely resemble the clean imagery (GT). These quantitative and qualitative results validate that even when temporal sequence context is unavailable (T = 1), the cloud-deterministic diffusion guidance in CDPCR effectively leverages spatial priors to restore high-resolution imagery.

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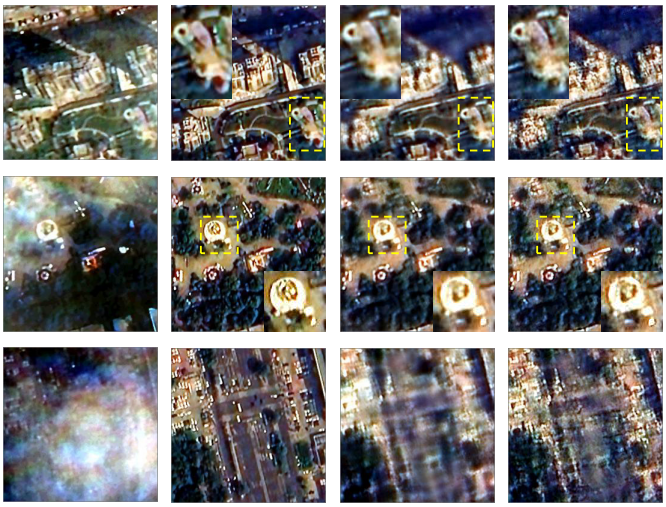


Fig. 10. Visualizations of reconstruction results on the high-resolution CUHK-CR dataset ($T = 1$). Specific subregions (yellow squares) are zoomed-in for better comparison.

C. Prospects for Diffusion-Based Cloud Removal Models

Future research could further refine CDPCR's unified workflow by incorporating fine-grained physical priors into the diffusion trajectory. Such advancements would enhance the interpretability and fidelity of diffusion-based reconstruction in real-world remote sensing applications.

Furthermore, we observe that monitoring high-frequency information contained in the recovered images helps to improve the sampling process (V-A4). Therefore, leveraging the unique frequency characteristics of clouds, integrating frequency-domain information into diffusion models remains a key focus for future research.

VI. Conclusion

Given unpredictable atmospheric conditions, natural cloud contamination exhibits high diversity and variability. Intermingled cloud layers of varying thickness are prevalent in practical scenarios. Such contamination, involving complex interactions (e.g., visibility reduction and shadow casting) with diverse ground features and land cover changes, severely limits conventional cloud removal methods. To address these limitations, we propose the cloud-deterministic diffusion model (CDPCR). Unlike standard diffusion paradigms, CDPCR introduces a novel cloud-deterministic forward process that models natural cloud contamination as the sole degradation source, providing the network with extensive exposure to diverse realistic cloud patterns and varying intensities during training. For inference, an improved iterative sampling algorithm that effectively reverses degradation is adopted to achieve robust S2S reconstruction for optical time series.

Extensive cross-dataset experiments demonstrate the effectiveness and robustness of CDPCR in handling complex cloud contamination across diverse surface types. The results indicate that CDPCR outperforms existing SOTA methods, reconstructing underlying features with stronger structural integrity and more realistic ground details. Notably, the model adaptively manages intermingled cloud layers without explicit classification, moving beyond the conventional binary "thin/thick" dichotomy to provide a more nuanced and automated understanding of natural scenarios. Systematic ablation analyses further validate the superiority and applicability of the proposed model from multiple perspectives. By mitigating the dependence on clean temporal references and external masks, CDPCR provides a highly flexible and efficient solution for cloud removal in real-world RSI.

Appendix A
Standard Sampling Algorithm

The following is the standard sampling algorithm (Algorithm 1) used in diffusion models [60].

Algorithm 1 Standard Sampling

Input: A degraded sample $\mathbf{X}_S$
1 **for** $s = S, S-1, \cdots, 1$ **do**
2 $\quad \widehat{\mathbf{X}}_0 \leftarrow \mathcal{R}(\mathbf{X}_s, s)$
3 $\quad \mathbf{X}_{s-1} = \mathcal{D}(\widehat{\mathbf{X}}_0, s-1)$
4 **end for**
Return: $\widehat{\mathbf{X}}_0$

Algorithm 2 Improved Sampling

Input: A degraded sample $\mathbf{X}_S$
1 **for** $s = S, S-1, \cdots, 1$ **do**
2 $\quad \widehat{\mathbf{X}}_0 \leftarrow \mathcal{R}(\mathbf{X}_s, s)$
3 $\quad \mathbf{X}_{s-1} = \mathbf{X}_s - \mathcal{D}(\widehat{\mathbf{X}}_0, s) + \mathcal{D}(\widehat{\mathbf{X}}_0, s-1)$
4 **end for**
Return: $\widehat{\mathbf{X}}_0$

Appendix B
Supplementary Experimental Results

A. Experimental Results on SEN12MS-CR-TS

1) Visualization of Reconstructions Across Varying Cloud Contamination Levels: Fig. 11 provides an overview of model performances across scenarios with varying cloud contamination levels (S010–S111). When clouds are small and thin [see Fig. 11(a)–(i), S010-, S101- and S102-type], most methods perform well, given the ample temporal information available in the observations. Fig. 11(g)–(i) displays cloud removal for observations featuring highly reflective surfaces (e.g., snow). In these challenging S102-type scenarios featuring snow coverage, CDPCR exhibits high sensitivity in spectral–temporal disambiguation, effectively retrieving ground details without misinterpreting highly reflective snow as cloud cover, as demonstrated by the sharper and more reliable results in Fig. 11(i). As cloud contamination intensifies [see Fig. 11(j)–(v), S110- and S111-type], differences in model performance become more apparent. By comparison, CDPCR effectively removes contamination to retrieve coherent structures and realistic ground details. These experiments systematically assess CDPCR's capability and robustness to effectively remove cloud and shadow contamination across varying cloud thickness, coverage, and distribution.

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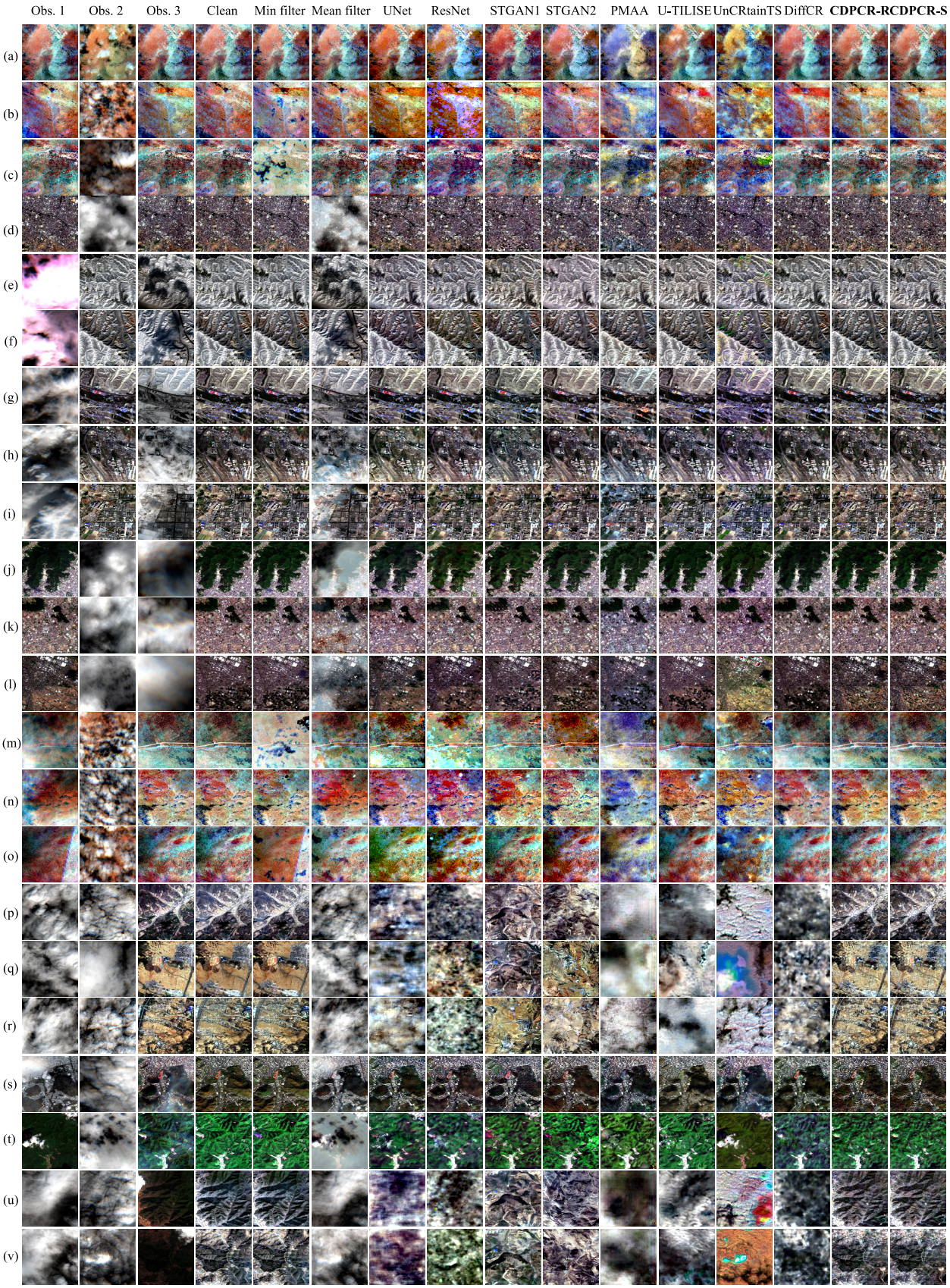


Fig. 11. Visual comparison of reconstruction results from different models on the SEN12MS-CR-TS dataset, showcasing the six distinct scenarios (from S010 to S111). The observed sequences feature diverse surface types and varying levels of cloud contamination. (a)–(v) Scenarios with varying levels of cloud contamination: (a)–(d) S010-type, (e)–(i) S101/S102-type, (j)–(l) S011-type, (m)–(r) S110-type, and (s)–(v) S111-type.

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